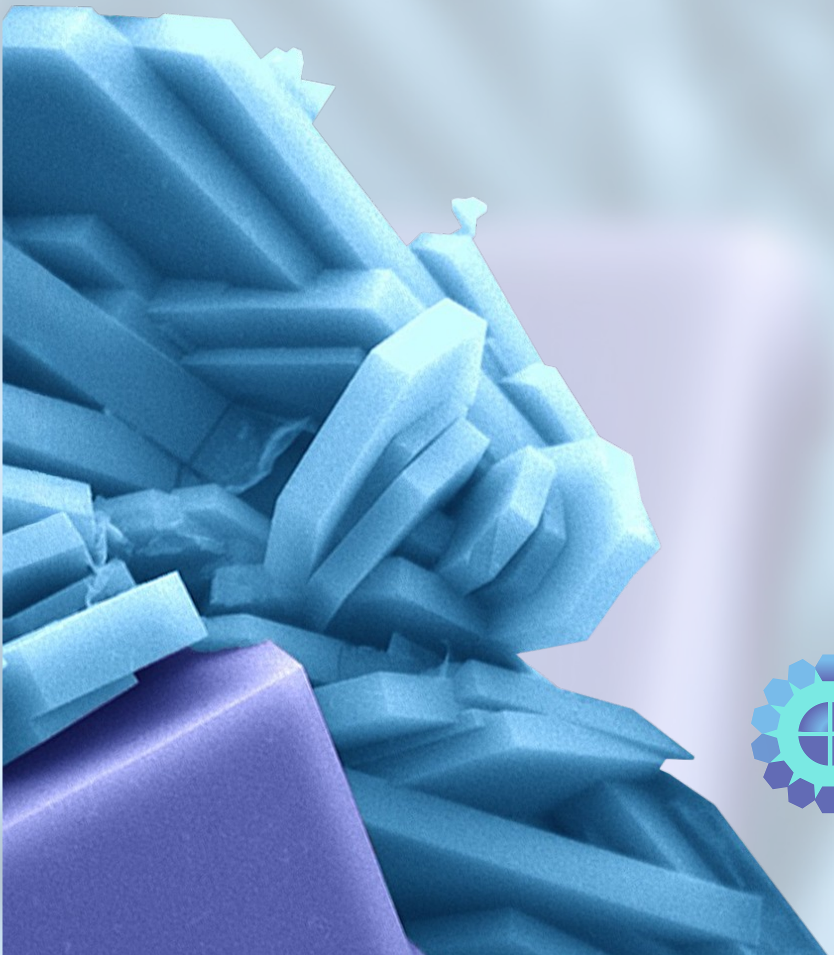

Molecular dynamics Hands-on session

Simon Gravelle
LIPhy (Grenoble France), CNRS

What you will learn today

- The very basics of molecular dynamics

To go further:

Understanding Molecular Simulation by Daan Frenkel and Berend Smit

https://lammpstutorials.github.io

- How to launch and modify a simple input

Large-scale Atomic/Molecular
Massively Parallel Simulator

Download LAMMPS-GUI

https://github.com/simongravelle/DIADEM

Download LAMMPS-GUI

https://github.com/simongravelle/DIADEM

LAMMPS files | DIADEM

Download LAMMPS-GUI

LAMMPS-GUI can be downloaded from https://github.com/akohlmey/lammps-gui/releases/tag/v3.0.7.

Alternatively, LAMMPS-GUI can also be downloaded from these links:

- LAMMPS-GUI (.exe) for Windows
- LAMMPS-GUI (.dmg) for macOS
- LAMMPS-GUI (.tar.gz) for Linux (tarball, preferred option)
- LAMMPS-GUI (.flatpak) for Linux (flatpak)

Download LAMMPS-GUI

https://github.com/simongravelle/DIADEM

LAMMPS files | DIADEM

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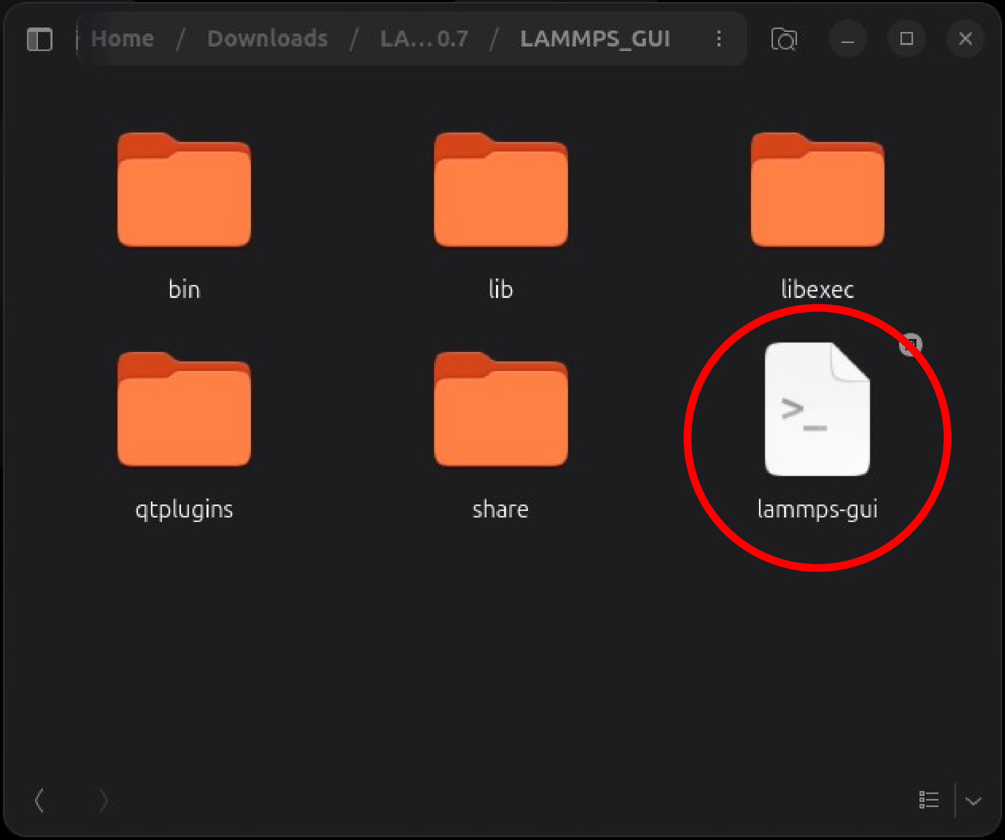
- LAMMPS-GUI (.flatpak) for Linux (flatpak)

https://lammps-gui.lammps.org/installation.html

Download LAMMPS-GUI

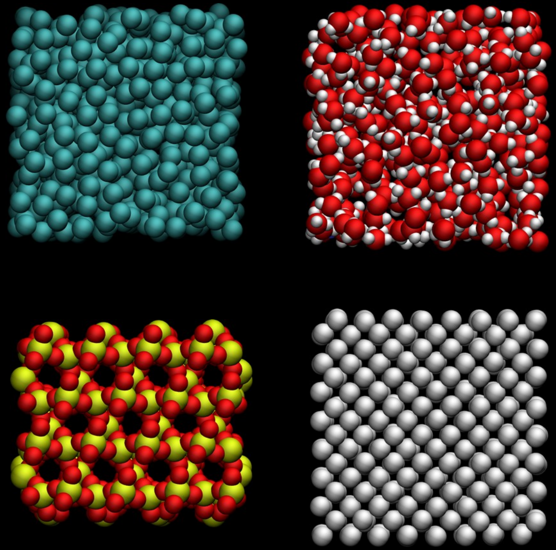


https://github.com/simongravelle/DIADEM

Explore the repository

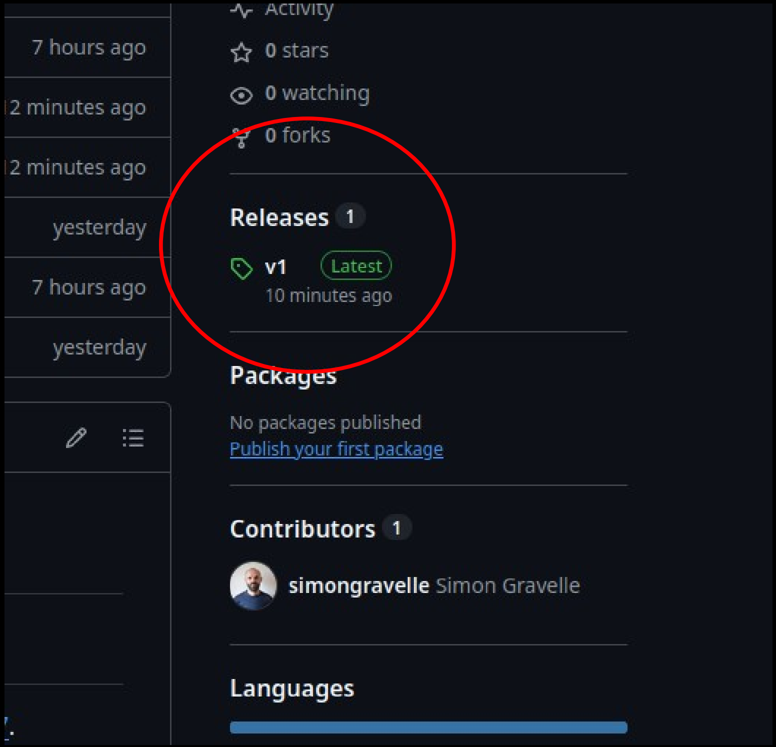
https://github.com/simongravelle/DIADEM

Explore the repository



https://github.com/simongravelle/DIADEM

Explore the repository

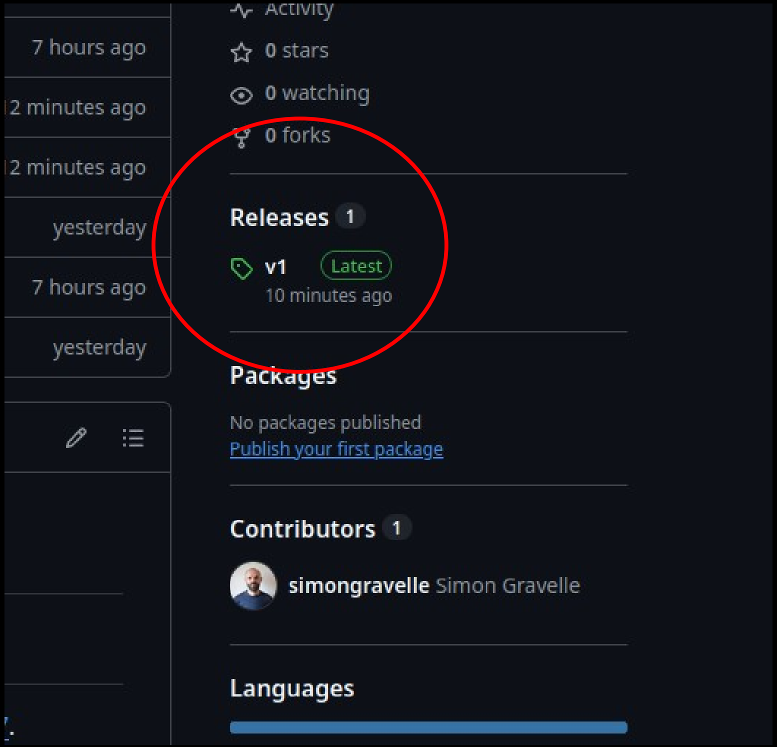


https://github.com/simongravelle/DIADEM

Explore the repository

https://tinyurl.com/y9w53mrj (zip)

https://tinyurl.com/56mecmb3 (tar.gz)



https://github.com/simongravelle/DIADEM


LAMMPS-GUI - Editor - *unknown*
File Edit Run View Tutorials About
1 # When using LAMMPS-GUI in your project, please cite: https://doi.org/10.33011/livecoms.6.1.3037
2
LAMMPS-GUI
Ready.
Directory: /home/simon/Git/Communication/LIPhy/2026/DIADEM/codes/argon-lj

First input

https://github.com/simongravelle/DIADEM

The first input can be downloaded from argon-lj/NVE/:

```
# Simple NVE argon simulation

units           lj # Use Lennard-Jones reduced units
dimension       3 # Perform the simulation in 3 spatial dimensions
atom_style      atomic # Atoms are treated as point particles without bonds or molecular topology
boundary        p p p # Use periodic boundary conditions

region          simbox block -20 20 -20 20 -20 20 # Define the simulation box from -20 to 20 in x, y, and z
create_box      1 simbox # Create a simulation box containing one atom type
create_atoms    1 random 600 34134 simbox overlap 0.7 # Randomly create N atoms inside the simulation box

mass            1    1.0 # Assign a mass to atom type 1 (mass = 1.0)
pair_style      lj/cut 4.0 # Use a Lennard-Jones pair potential with a cutoff distance of 4.0
pair_coeff      1    1    1.0 1.0 # Set Lennard-Jones parameters for atom type 1 (epsilon = 1.0 sigma = 1.0)

fix             mynve    all      nve # Integrate the equations of motion
timestep        0.005 # Set the integration timestep to 0.005 in LJ reduced time units

thermo          10 # Print thermodynamic information in log
thermo_style    custom step temp etotal ke pe # Choose what information is printed

# Uncomment to print atom coordinate to file
# dump            mydmp    all      custom 500 dump.lammpstrj id type x y z # Write atom coordinates to file

# Uncomment to generate images
# dump            viz      all      image 500 myimage-*.ppm type type size 800 800 zoom 1.452 shiny 0.5 fsaa yes view
# dump_modify     viz pad 9 boxcolor white backcolor black adiam 1 2 acolor 1 cyan

run             25000
```

LAMMPS-GUI - Editor - input.lmp

File Edit Run View Tutorials About

```
# Simple NVE argon simulation

units           lj # Use Lennard-Jones reduced units
dimension       3 # Perform the simulation in 3 spatial dimensions
atom_style      atomic # Atoms are treated as point particles without bonds or molecular topology
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# Uncomment to generate images
dump            viz      all      image 500 myimage-*.ppm type type size 800 800 zoom 1.452 shiny 0.5 fsaa yes view 0 0 box yes 0.005 axes no 0.0 0.0 center s 0.483725 0.510373 0.510373
dump_modify     viz pad 9 boxcolor white backcolor black adiam 1 2 acolor 1 cyan

run             25000

```
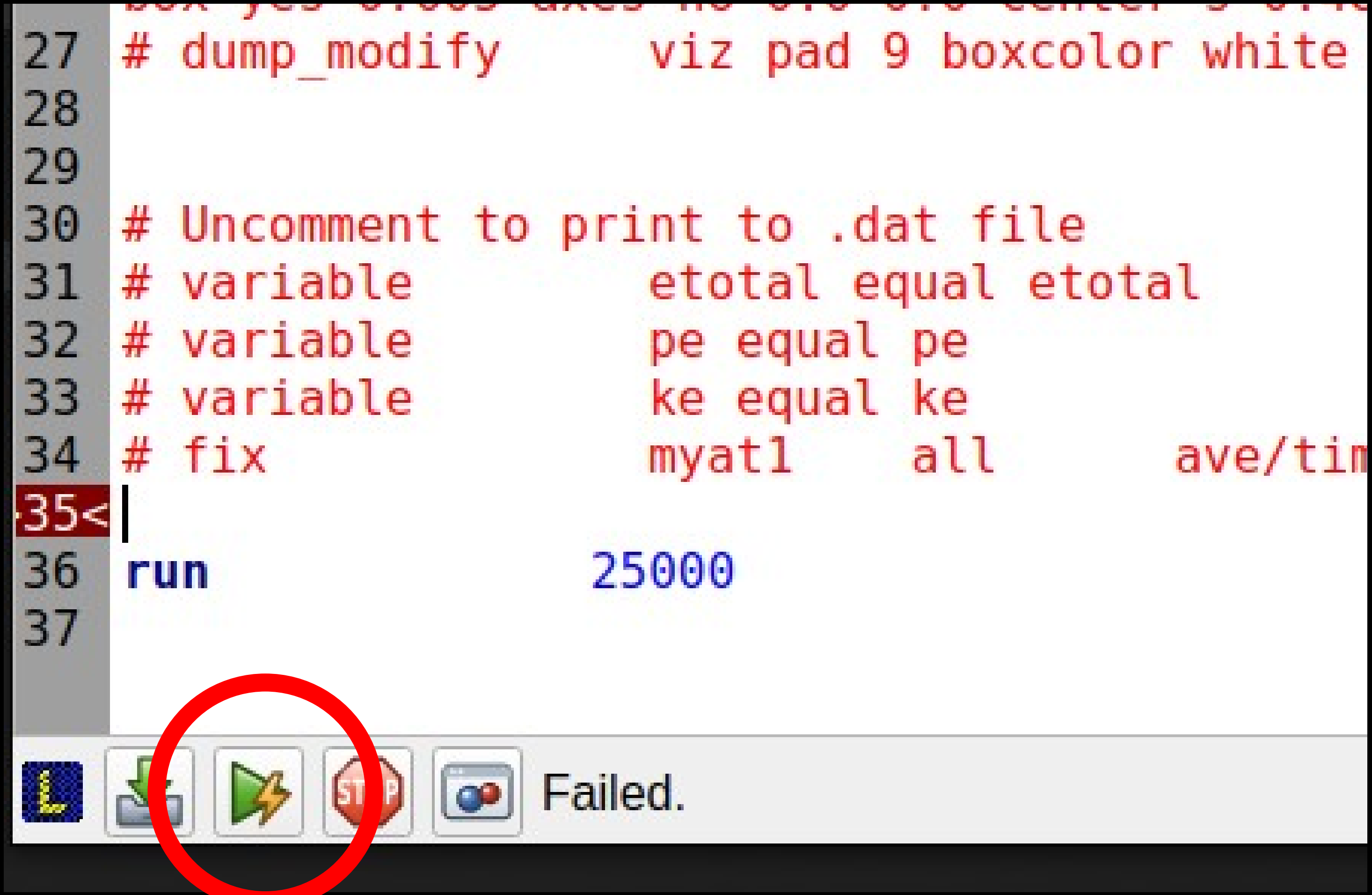
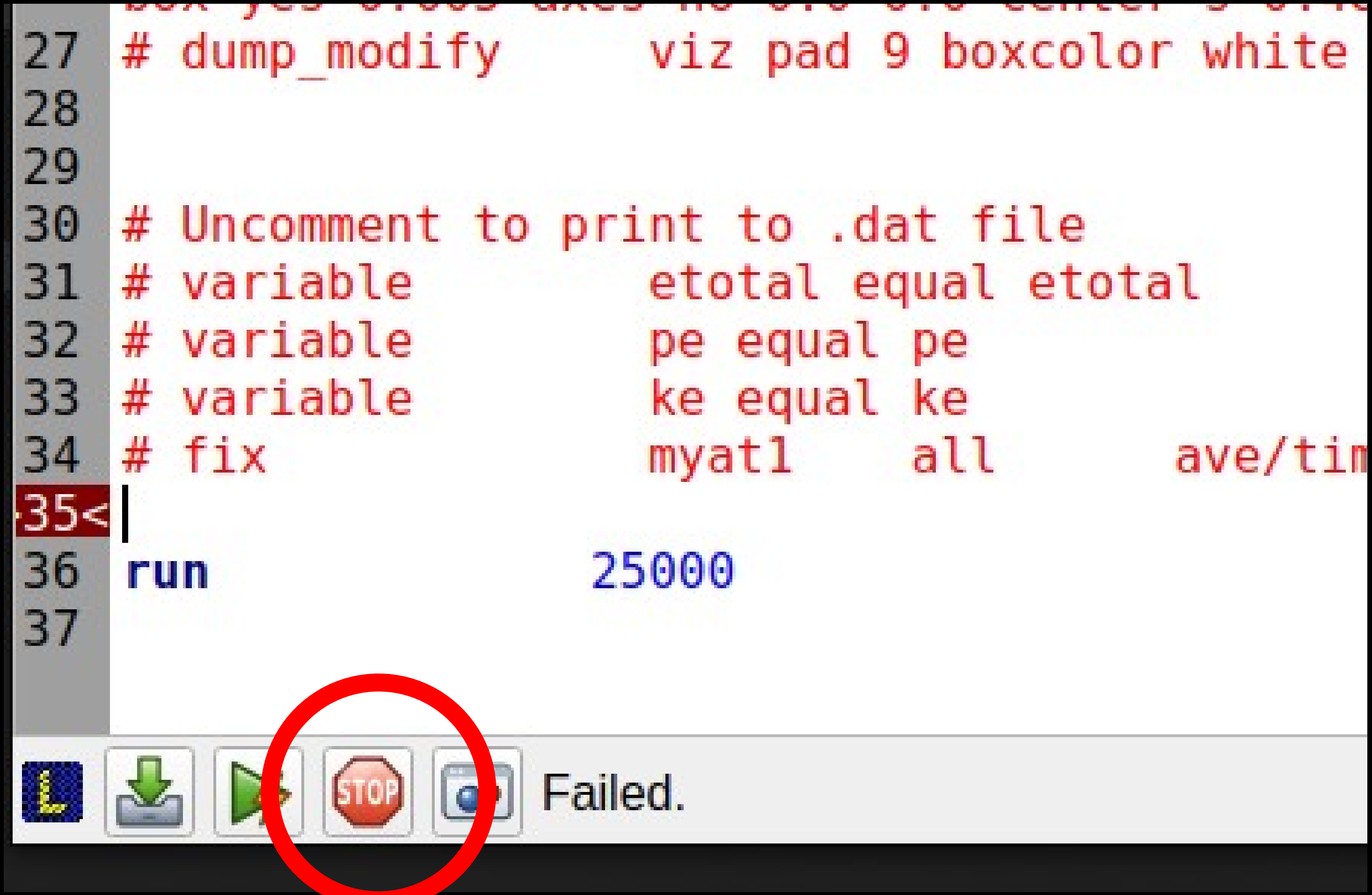

Ready. Directory: /home/simon/Git/Communication/LIPhy/2026/DIADEM/codes/argon-lj

```
# dump_modify     viz pad 9 boxcolor white

# Uncomment to print to .dat file
# variable        etotal equal etotal
# variable        pe equal pe
# variable        ke equal ke
# fix             myat1    all      ave/tim

run              25000
```

Failed.



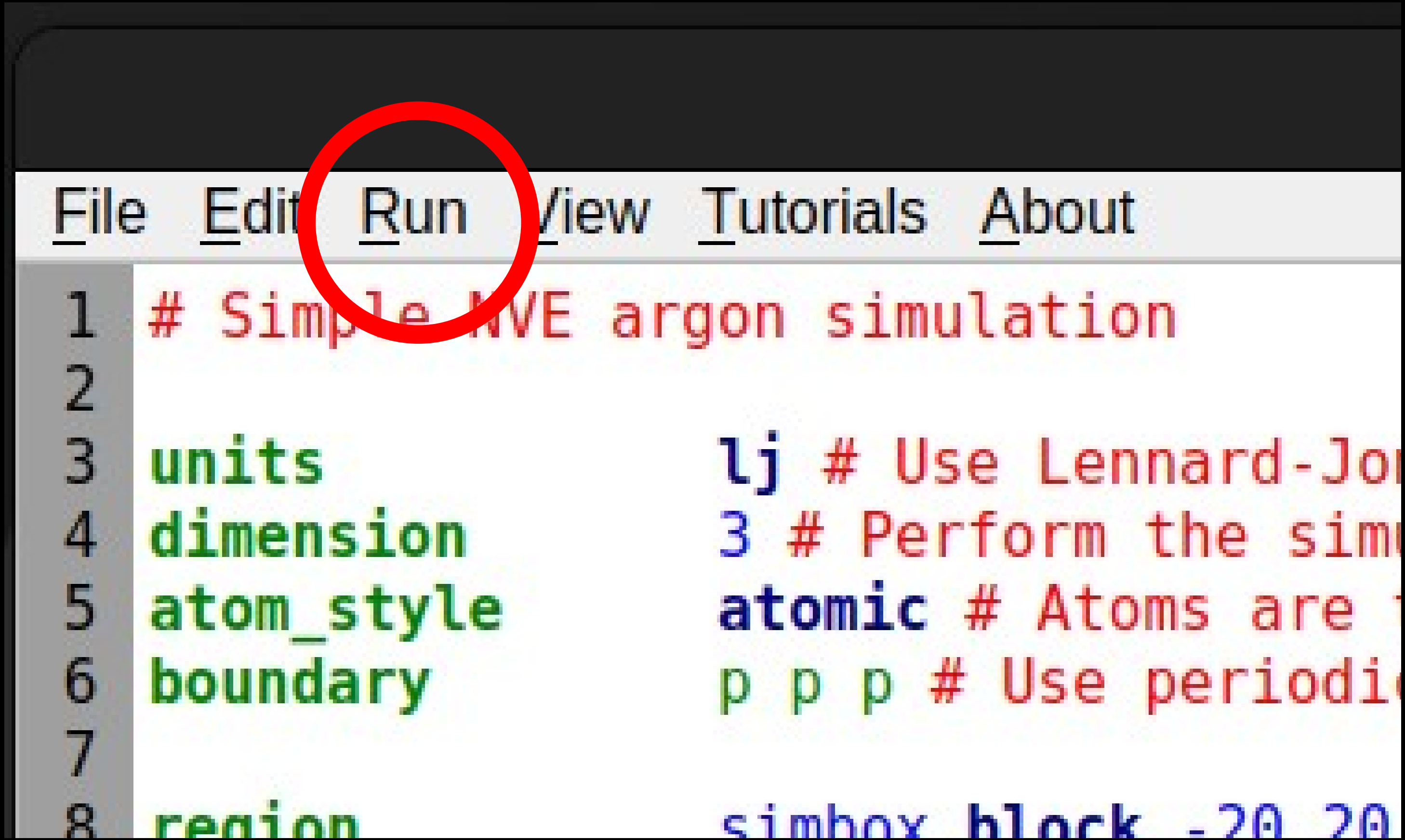
dump_modify viz pad 9 boxcolor white
Uncomment to print to .dat file
variable etotal equal etotal
variable pe equal pe
variable ke equal ke
fix myat1 all ave/tim
run 25000
Failed.



dump_modify viz pad 9 boxcolor white
Uncomment to print to .dat file
variable etotal equal etotal
variable pe equal pe
variable ke equal ke
fix myat1 all ave/tim
run 25000
STOP
Failed.

File Edit Run View Tutorials About

```
# Simple NVE argon simulation

units               lj # Use Lennard-Jo
dimension           3 # Perform the simu
atom_style          atomic # Atoms are 
boundary            p p p # Use periodic

region              simbox block -20 20
```


File Edit Run View Tutorials About
1 # Simple NVE argon simulation
2
3 units lj # Use Lennard-Jo
4 dimension 3 # Perform the sim
5 atom_style atomic # Atoms are
6 boundary p p p # Use periodi
7
8 region simbox block -20 20

LAMMPS-GUI - Editor - input.lmp

File Edit Run View Tutorials About

```
# Simple NVE argon simulation

units           lj # Use Lennard-Jones reduced units
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pair_coeff      1    1    1.0 1.0 # Set Lennard-Jones parameters for atom type 1 (epsilon = 1.0 sigma = 1.0)

fix             mynve    all      nve # Integrate the equations of motion
timestep        0.005 # Set the integration timestep to 0.005 in LJ reduced time units

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# Uncomment to print atom coordinate to file
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# Uncomment to generate images
dump            viz      all      image 500 myimage-*.ppm type type size 800 800 zoom 1.452 shiny 0.5 fsaa yes view 0 0 box yes 0.005 axes no 0.0 0.0 center s 0.483725 0.510373 0.510373
dump_modify     viz pad 9 boxcolor white backcolor black adiam 1 2 acolor 1 cyan

run             25000

```

Ready. Directory: /home/simon/Git/Communication/LIPhy/2026/DIADEM/codes/argon-lj

LAMMPS commands

```
units           lj # Use Lennard-Jones reduced units
dimension       3 # Perform the simulation in 3 spatial dimensions
atom_style      atomic # Atoms are treated as point particles without bonds or molecular topology
boundary        p p p # Use periodic boundary conditions
```

LAMMPS commands

```
units           lj # Use Lennard-Jones reduced units
dimension       3 # Perform the simulation in 3 spatial dimensions
atom_style      atomic # Atoms are treated as point particles without bonds or molecular topology
boundary        p p p # Use periodic boundary conditions
```

Each line is a command

LAMMPS commands

```
units           lj # Use Lennard-Jones reduced units
dimension       3 # Perform the simulation in 3 spatial dimensions
atom_style      atomic # Atoms are treated as point particles without bonds or molecular topology
boundary        p p p # Use periodic boundary conditions
```

Each line is a command

Documentation available by right click (“View Documentation for ...”)

LAMMPS commands

```
units           lj # Use Lennard-Jones reduced units
dimension       3 # Perform the simulation in 3 spatial dimensions
atom_style      atomic # Atoms are treated as point particles without bonds or molecular topology
boundary        p p p # Use periodic boundary conditions
```

units command

Syntax

```
units style
```

- style = *lj* or *real* or *metal* or *si* or *cgs* or *electron* or *micro* or *nano*

Examples

```
units metal
units lj
```

Description

This command sets the style of units used for a simulation. It determines the units of all quantities specified in the input script and data file, as well as quantities output to the screen, log file, and dump files. Typically, this command is used at the very beginning of an input script.

For all units except *lj*, LAMMPS uses physical constants from www.physics.nist.gov. For the definition of kcal in real units, LAMMPS uses the thermochemical calorie = 4.184 J.

LAMMPS commands

```
unit            lj # Use Lennard-Jones reduced units
dimension       3 # Perform the simulation in 3 spatial dimensions
atom_style      atomic # Atoms are treated as point particles without bonds or molecular topology
boundary        p p p # Use periodic boundary conditions
```

Each line is a command

Documentation available by right click ("View Documentation for ...")

LAMMPS commands

```
unit            lj # Use Lennard-Jones reduced units
dimension       3 # Perform the simulation in 3 spatial dimensions
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Each line is a command

Documentation available by right click (“View Documentation for ...”)

The GUI provides color coding and autocomplete

LAMMPS commands

```
units           lj # Use Lennard-Jones reduced units
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LAMMPS commands

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units           lj # Use Lennard-Jones reduced units
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atom_style      atomic # Atoms are treated as point particles without bonds or molecular topology
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```

x y z

Each line is a command

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LAMMPS commands

```
units           lj # Use Lennard-Jones reduced units
dimension       3 # Perform the simulation in 3 spatial dimensions
atom_style      atomic # Atoms are treated as point particles without bonds or molecular topology
boundary        p p p # Use periodic boundary conditions
```

Each line is a command

Documentation available by right click (“View Documentation for ...”)

The GUI provides color coding and autocomplete

Everything after # is a comment

LAMMPS commands

```
region          simbox block -20 20 -20 20 -20 20 # Define the simulation box from -20 to 20 in x, y, and z
create_box      1 simbox # Create a simulation box containing one atom type
create_atoms    1 random 600 34134 simbox overlap 0.7 # Randomly create N atoms inside the simulation box
```

LAMMPS commands

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region           simbox block -20 20 -20 20 -20 20 # Define the simulation box from -20 to 20 in x, y, and z
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```

LAMMPS commands

x y z

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create_box      1 simbox # Create a simulation box containing one atom type
create_atoms    1 random 600 34134 simbox overlap 0.7 # Randomly create N atoms inside the simulation box
```

LAMMPS commands

One atom type

```
region          simbox block -20 20 -20 20 -20 20 # Define the simulation box from -20 to 20 in x, y, and z
create_box      1 simbox # Create a simulation box containing one atom type
create_atoms    1 random 600 34134 simbox overlap 0.7 # Randomly create N atoms inside the simulation box
```

LAMMPS commands

```
region          simbox block -20 20 -20 20 -20 20 # Define the simulation box from -20 to 20 in x, y, and z
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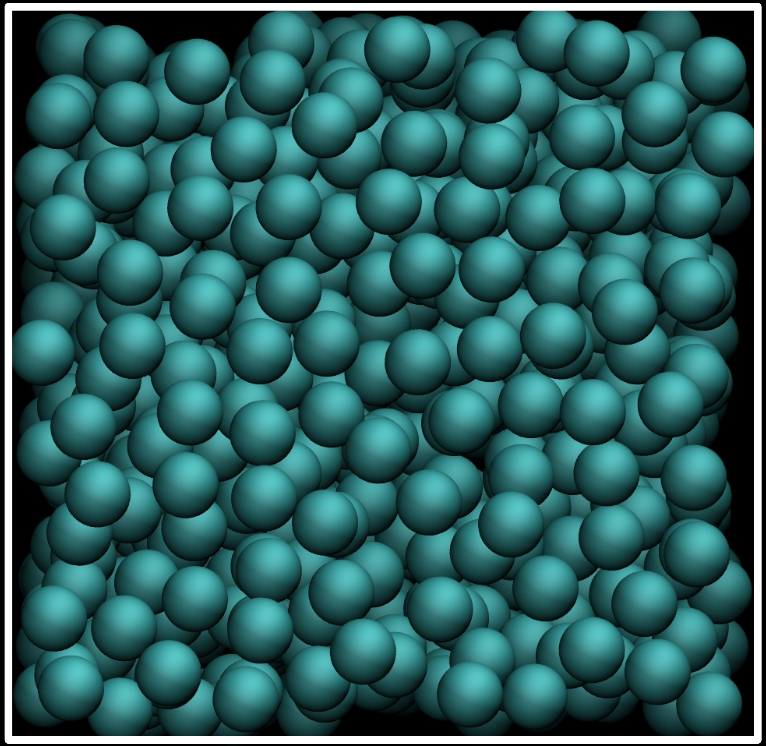

LAMMPS commands

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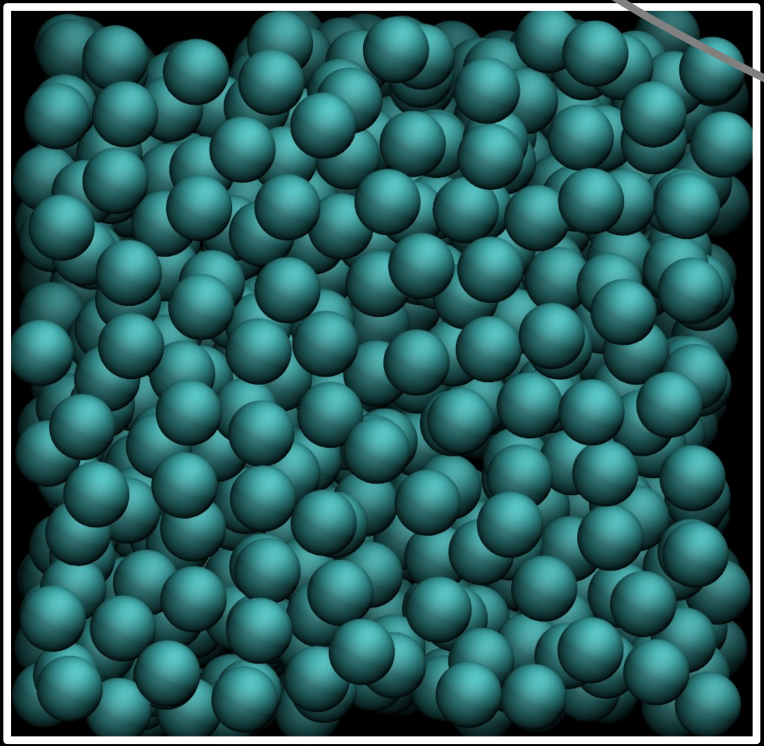

Number of atoms

LAMMPS commands

```
region          simbox block -20 20 -20 20 -20 20 # Define the simulation box from -20 to 20 in x, y, and z
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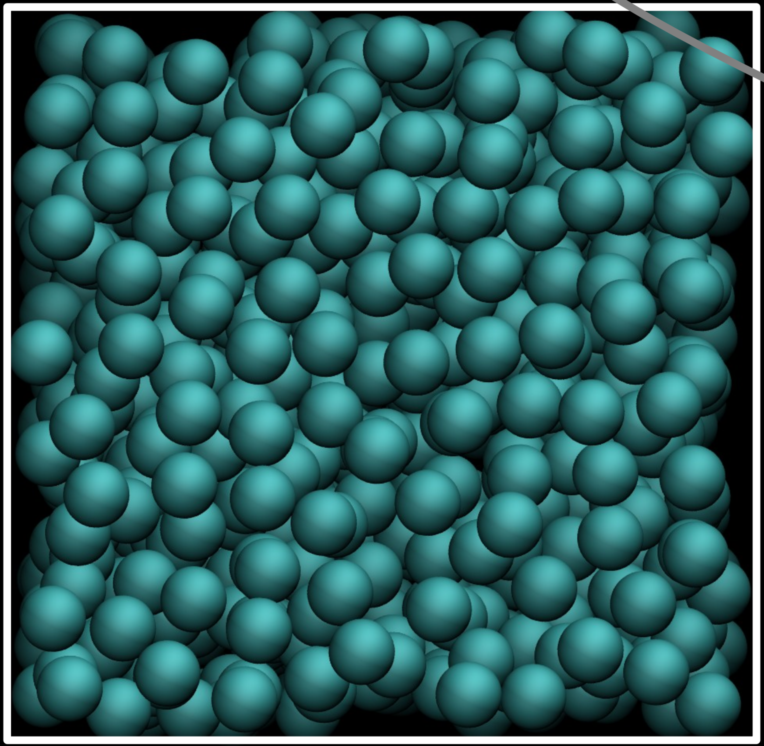

Don't create atoms closer than 0.7

Number of atoms

LAMMPS commands

```
mass            1    1.0 # Assign a mass to atom type 1 (mass = 1.0)
pair_style      lj/cut 4.0 # Use a Lennard-Jones pair potential with a cutoff distance of 4.0
pair_coeff      1    1    1.0 1.0 # Set Lennard-Jones parameters for atom type 1 (epsilon = 1.0 sigma = 1.0)
```

LAMMPS commands

```
mass            1    1.0 # Assign a mass to atom type 1 (mass = 1.0)
pair_style      lj/cut 4.0 # Use a Lennard-Jones pair potential with a cutoff distance of 4.0
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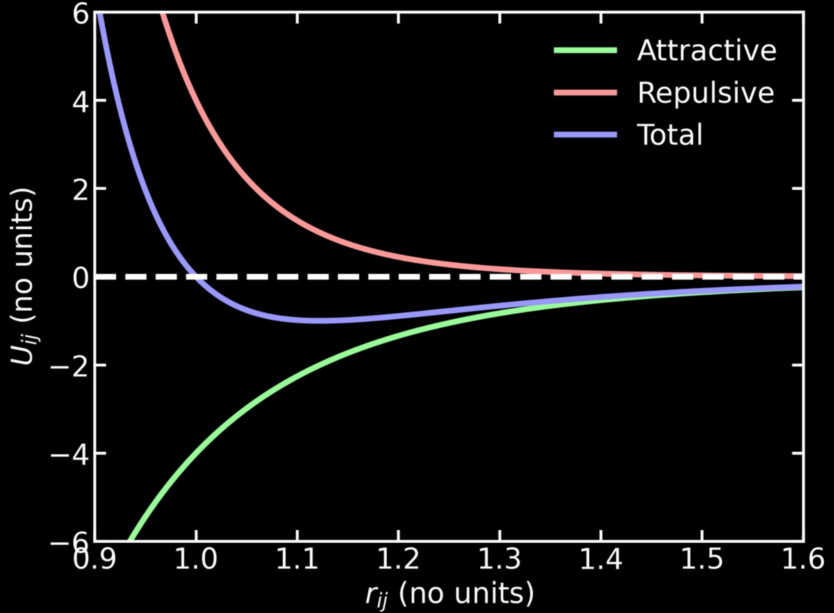
```



$$U_{ij}(r_{ij}) = 4\varepsilon_{ij}\left[\left(\frac{\sigma_{ij}}{r_{ij}}\right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}}\right)^{6}\right]$$

σ_{ij} : effective particle size

ϵ_{ij} : interaction strength

LAMMPS commands

```
fix             mynve    all      nve # Integrate the equations of motion
timestep        0.005 # Set the integration timestep to 0.005 in LJ reduced time units

thermo          10 # Print thermodynamic information in log
thermo_style    custom step temp etotal ke pe # Choose what information is printed
```

LAMMPS commands

```
fix             mynve    all      nve # Integrate the equations of motion
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thermo          10 # Print thermodynamic information in log
thermo_style    custom step temp etotal ke pe # Choose what information is printed
```

$$\sum_{j \neq i} \mathbf{F}_{ij} = m_i \ddot{\mathbf{r}}_i$$

Verlet algorithm

$$\mathbf{r}(t + \Delta t) = \mathbf{r}(t) + \dot{\mathbf{r}}(t)\Delta t + \frac{1}{2}\ddot{\mathbf{r}}(t)\Delta t^2$$

$$\dot{\mathbf{r}}(t + \Delta t) = \dot{\mathbf{r}}(t) + \frac{\ddot{\mathbf{r}}(t) + \ddot{\mathbf{r}}(t + \Delta t)}{2}\Delta t$$

LAMMPS commands

```
fix             mynve    all      nve # Integrate the equations of motion
timestep        0.005 # Set the integration timestep to 0.005 in LJ reduced time units

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LAMMPS commands

```
fix             mynve    all      nve # Integrate the equations of motion
timestep        0.005 # Set the integration timestep to 0.005 in LJ reduced time units

thermo          10 # Print thermodynamic information in log
thermo_style    custom step temp etotal ke pe # Choose what information is printed
```

Temperature

Potential energy

Total energy

Kinetic energy

LAMMPS commands

```
fix             mynve    all      nve # Integrate the equations of motion
timestep        0.005 # Set the integration timestep to 0.005 in LJ reduced time units

thermo          10 # Print thermodynamic information in log
thermo_style    custom step temp etotal ke pe density # Choose what information is printed
```

```
possible keywords = step, elapsed, elaplong, dt, time,
                    cpu, tpcpu, spcpu, cpuuse, cpuremain, part, timeremain,
                    atoms, temp, press, pe, ke, etotal,
                    evdwl, ecoul, epair, ebond, eangle, edihed, eimp,
                    emol, elong, etail,
                    enthalpy, ecouple, econserve,
                    vol, density,
                    xlo, xhi, ylo, yhi, zlo, zhi,
                    xy, xz, yz,
                    avecx, avecy, avecz,
                    bvecx, bvecy, bvecz,
                    cvecx, cvecy, cvecz,
                    lx, ly, lz,
                    xlat, ylat, zlat,
                    cella, cellb, cellc, cellalpha, cellbeta, cellgamma,
                    pxx, pyy, pzz, pxy, pxz, pyz,
                    bonds, angles, dihedrals, impropers,
                    fmax, fnorm, nbuild, ndanger,
                    c_ID, c_ID[I], c_ID[I][J],
                    f_ID, f_ID[I], f_ID[I][J],
                    v_name, v_name[I]
```

LAMMPS commands

```
fix            mynve    all      nve # Integrate the equations of motion
timestep       0.005 # Set the integration timestep to 0.005 in LJ reduced time units

thermo         10 # Print thermodynamic information in log
thermo_style   custom step temp etotal ke pe density # Choose what information is printed
```

```
possible keywords = step, elapsed, elaplong, dt, time,
                    cpu, tpcpu, spcpu, cpuuse, cpuremain, part, timeremain,
                    atoms, temp, press, pe, ke, etotal,
                    evdwl, ecoul, epair, ebond, eangle, edihed, eimp,
                    emol, elong, etail,
                    enthalpy, ecouple, econserve,
                    vol, density,
                    xlo, xhi, ylo, yhi, zlo, zhi,
                    xy, xz, yz,
                    avecx, avecy, avecz,
                    bvecx, bvecy, bvecz,
                    cvecx, cvecy, cvecz,
                    lx, ly, lz,
                    xlat, ylat, zlat,
                    cella, cellb, cellc, cellalpha, cellbeta, cellgamma,
                    pxx, pyy, pzz, pxy, pxz, pyz,
                    bonds, angles, dihedrals, impropers,
                    fmax, fnorm, nbuild, ndanger,
                    c_ID, c_ID[I], c_ID[I][J],
                    f_ID, f_ID[I], f_ID[I][J],
                    v_name, v_name[I]
```

LAMMPS commands

```
run             25000
```

Run simulation for 25000 steps

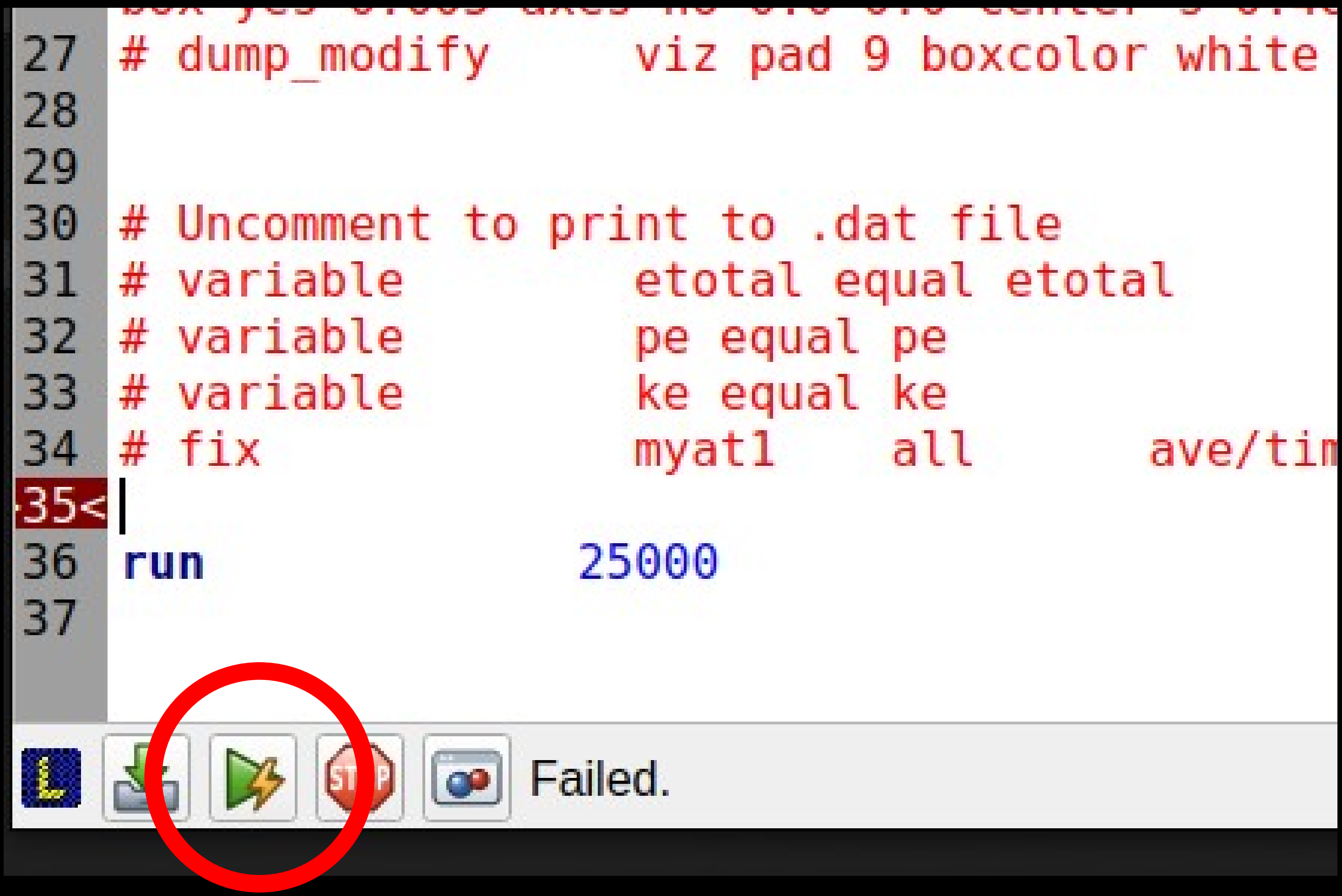
LAMMPS commands

```
fix             mynve    all       nve # Integrate the equations of motion
timestep        0.005 # Set the integration timestep to 0.005 in LJ reduced time units
```

```
run             25000
```

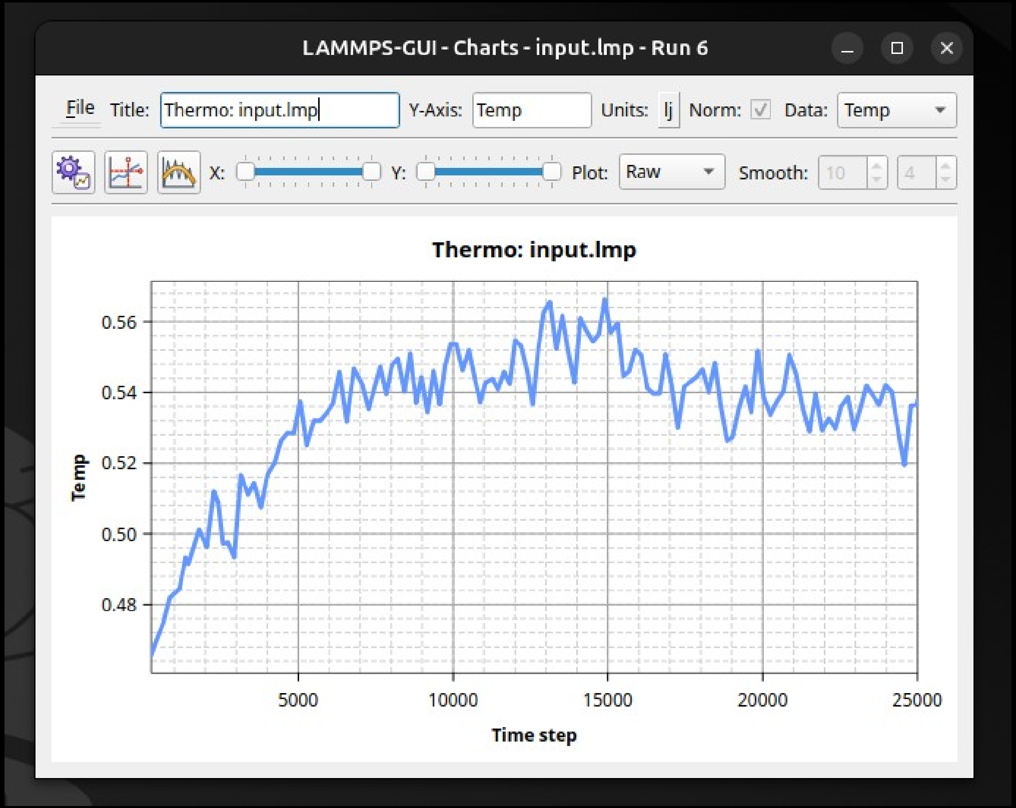
Run simulation for 25000 steps

With a timestep of 0.005, the total time is 125 (unitless)

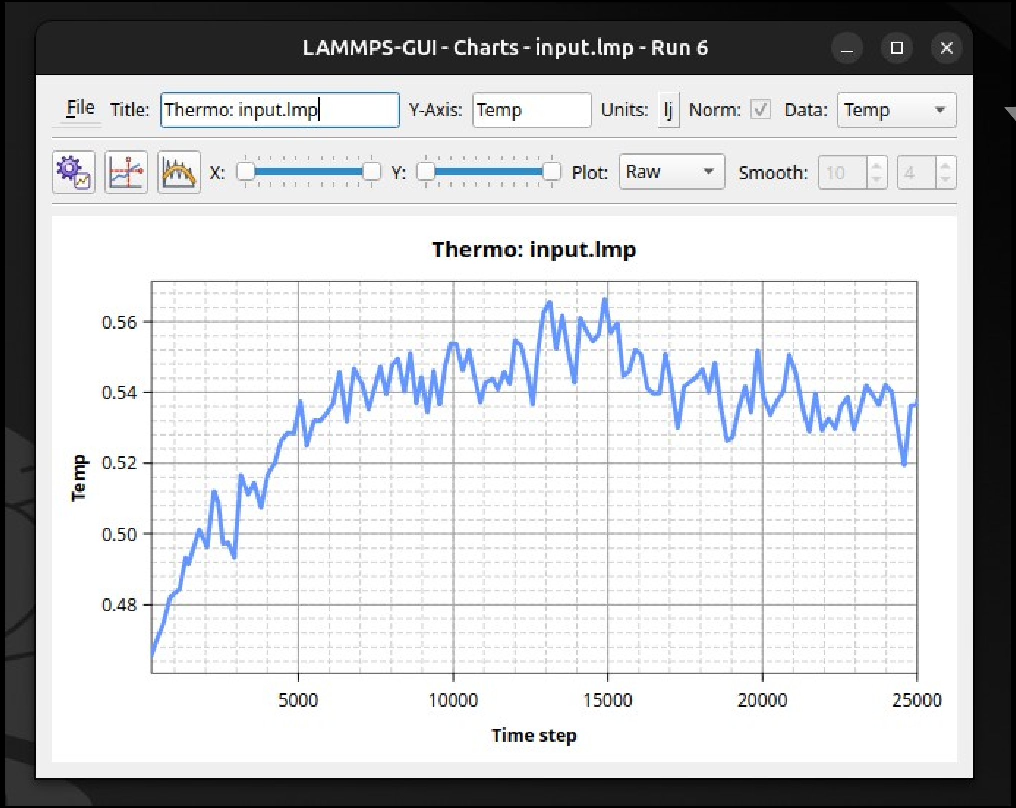


dump_modify viz pad 9 boxcolor white
Uncomment to print to .dat file
variable etotal equal etotal
variable pe equal pe
variable ke equal ke
fix myat1 all ave/tim
run 25000
Failed.

Select view → Charts window

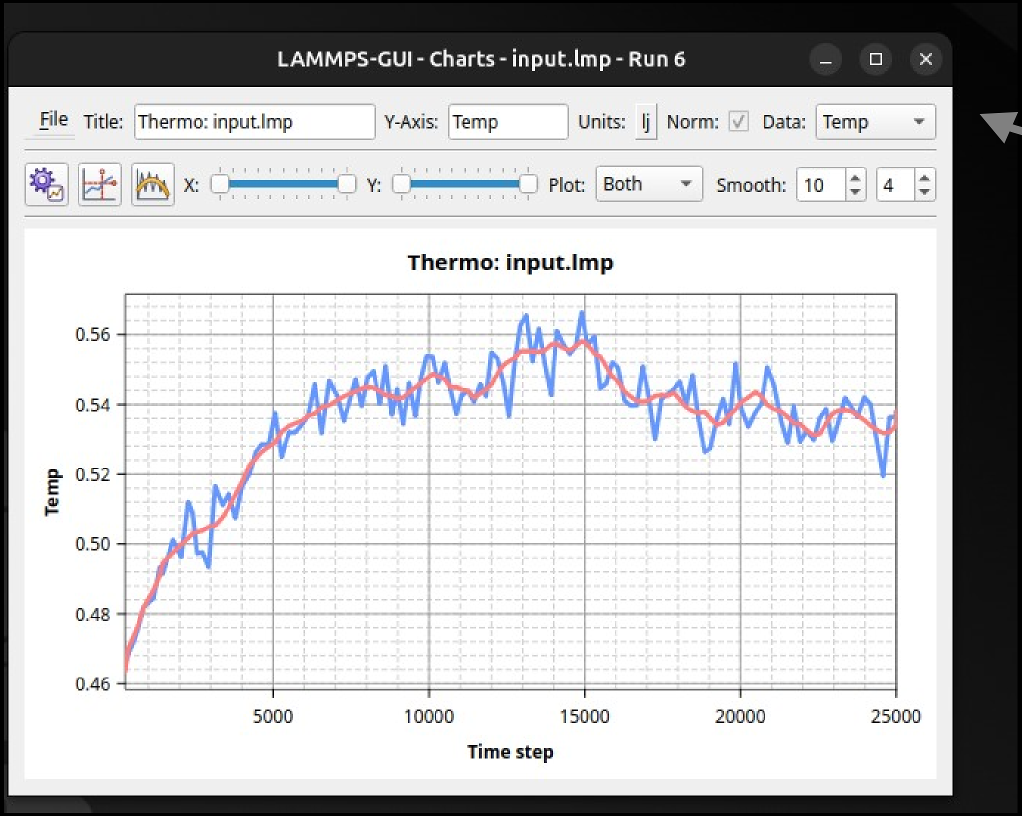


Select view → Charts window



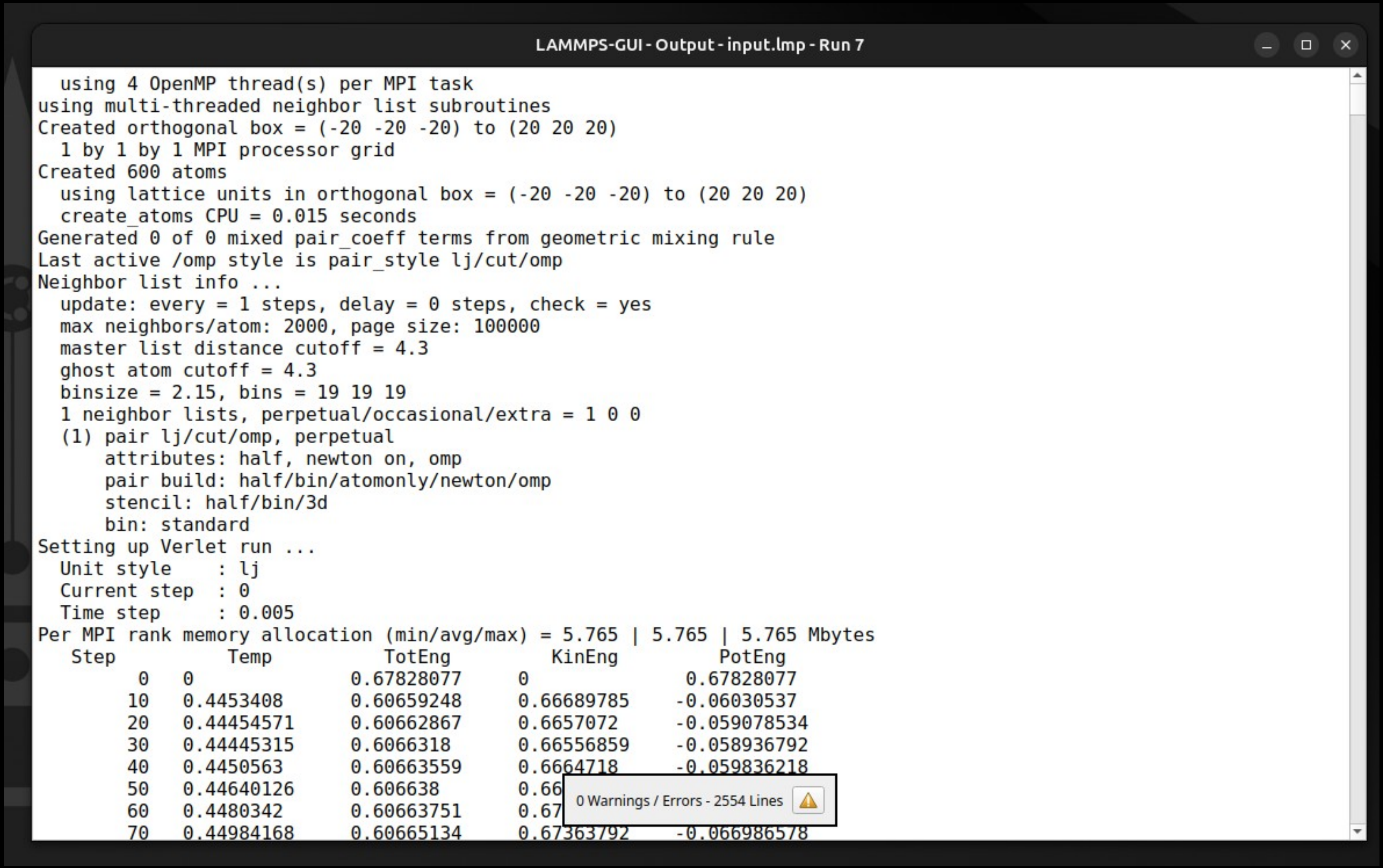
Select quantity here

Select view → Charts window



Select quantity here

Select view → Output window



Select view → Output window

LAMMPS-GUI - Output - input.lmp - Run 7

```
  using 4 OpenMP thread(s) per MPI task
using multi-threaded neighbor list subroutines
Created orthogonal box = (-20 -20 -20) to (20 20 20)
  1 by 1 by 1 MPI processor grid
Created 600 atoms
  using lattice units in orthogonal box = (-20 -20 -20) to (20 20 20)
  create_atoms CPU = 0.015 seconds
Generated 0 of 0 mixed pair_coeff terms from geometric mixing rule
Last active /omp style is pair_style lj/cut/omp
Neighbor list info ...
  update: every = 1 steps, delay = 0 steps, check = yes
  max neighbors/atom: 2000, page size: 100000
  master list distance cutoff = 4.3
  ghost atom cutoff = 4.3
  binsize = 2.15, bins = 19 19 19
  1 neighbor lists, perpetual/occasional/extra = 1 0 0
  (1) pair lj/cut/omp, perpetual
      attributes: half, newton on, omp
      pair build: half/bin/atomonly/newton/omp
      stencil: half/bin/3d
      bin: standard
Setting up Verlet run ...
  Unit style    : lj
  Current step  : 0
  Time step     : 0.005
Per MPI rank memory allocation (min/avg/max) = 5.765 | 5.765 | 5.765 Mbytes
   Step          Temp          TotEng         KinEng          PotEng
         0   0              0.67828077     0               0.67828077
        10   0.4453408      0.60659248     0.66689785     -0.06030537
        20   0.44454571     0.60662867     0.6657072      -0.059078534
        30   0.44445315     0.6066318      0.66556859     -0.058936792
        40   0.4450563      0.60663559     0.6664718      -0.059836218
        50   0.44640126     0.606638       0.66
        60   0.4480342      0.60663751     0.67
        70   0.44984168     0.60665134     0.67363792     -0.066986578
```

0 Warnings / Errors - 2554 Lines

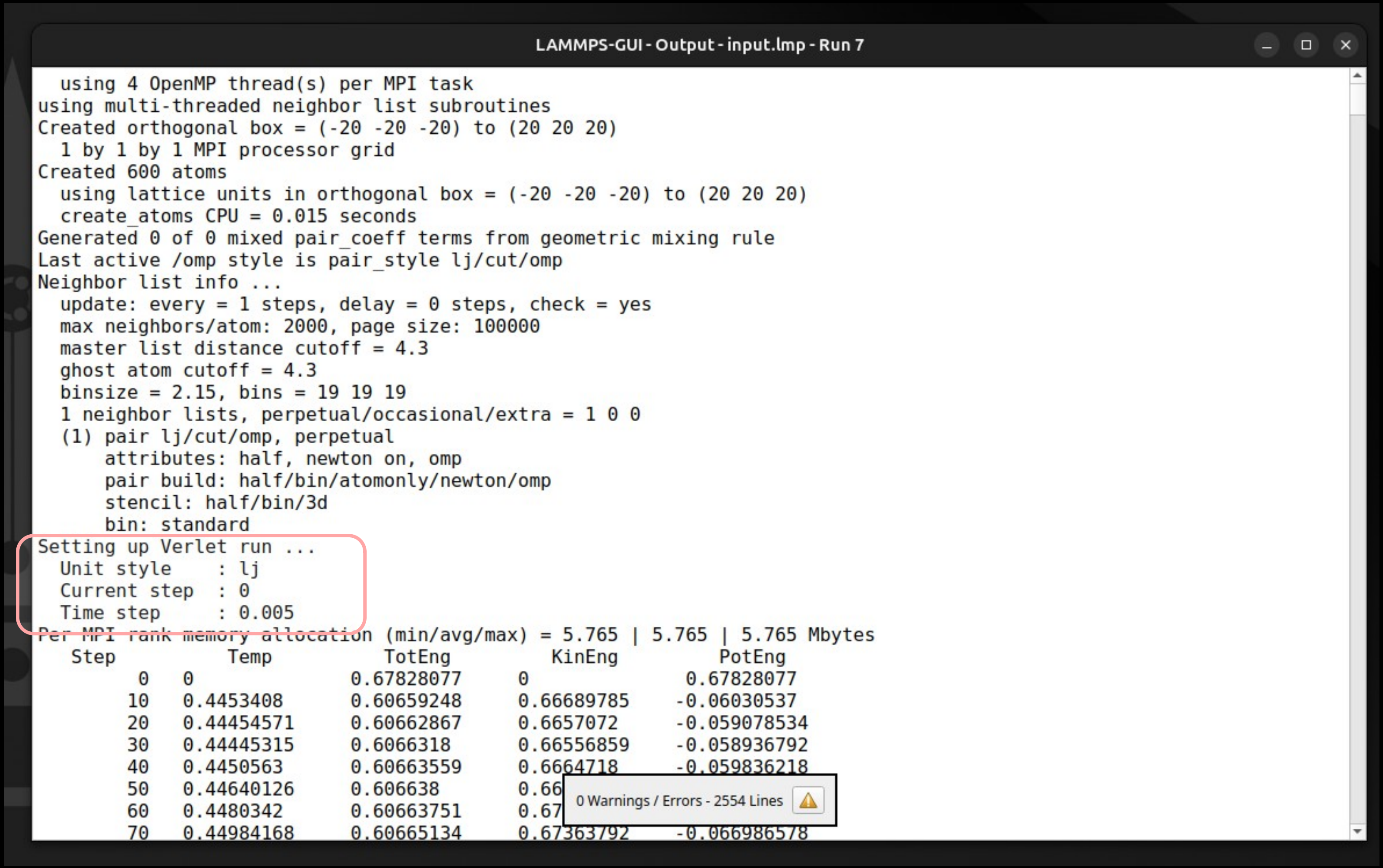
Select view → Output window

LAMMPS-GUI - Output - input.lmp - Run 7

```
  using 4 OpenMP thread(s) per MPI task
using multi-threaded neighbor list subroutines
Created orthogonal box = (-20 -20 -20) to (20 20 20)
  1 by 1 by 1 MPI processor grid
Created 600 atoms
  using lattice units in orthogonal box = (-20 -20 -20) to (20 20 20)
  create_atoms CPU = 0.015 seconds
Generated 0 of 0 mixed pair_coeff terms from geometric mixing rule
Last active /omp style is pair_style lj/cut/omp
Neighbor list info ...
  update: every = 1 steps, delay = 0 steps, check = yes
  max neighbors/atom: 2000, page size: 100000
  master list distance cutoff = 4.3
  ghost atom cutoff = 4.3
  binsize = 2.15, bins = 19 19 19
  1 neighbor lists, perpetual/occasional/extra = 1 0 0
  (1) pair lj/cut/omp, perpetual
      attributes: half, newton on, omp
      pair build: half/bin/atomonly/newton/omp
      stencil: half/bin/3d
      bin: standard
Setting up Verlet run ...
  Unit style    : lj
  Current step  : 0
  Time step     : 0.005
Per MPI rank memory allocation (min/avg/max) = 5.765 | 5.765 | 5.765 Mbytes
   Step          Temp          TotEng         KinEng         PotEng
         0   0              0.67828077     0              0.67828077
        10   0.4453408      0.60659248     0.66689785    -0.06030537
        20   0.44454571     0.60662867     0.6657072     -0.059078534
        30   0.44445315     0.6066318      0.66556859    -0.058936792
        40   0.4450563      0.60663559     0.6664718     -0.059836218
        50   0.44640126     0.606638       0.66
        60   0.4480342      0.60663751     0.67
        70   0.44984168     0.60665134     0.67363792    -0.066986578
```

0 Warnings / Errors - 2554 Lines

Select view → Output window



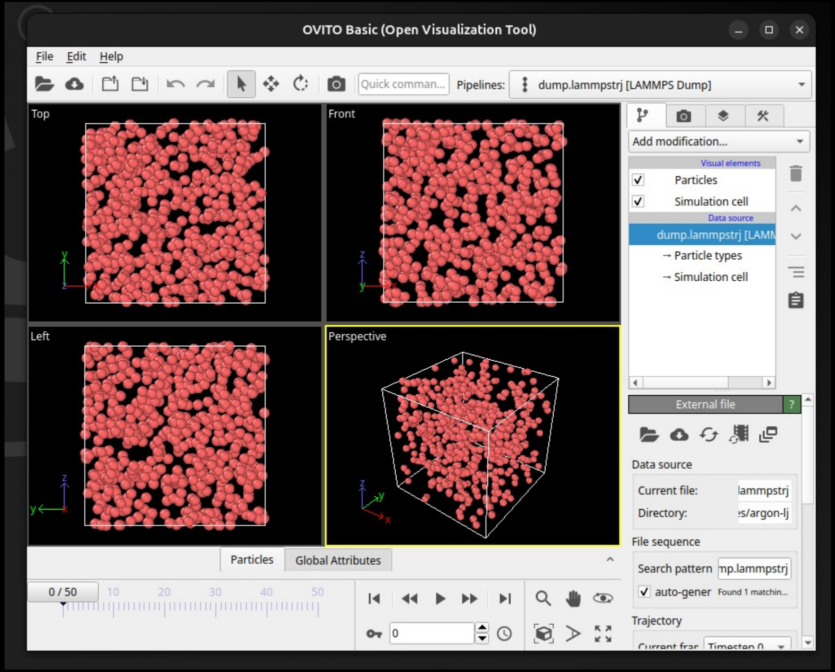
Visualization

```
# Uncomment to print atom coordinate to file
dump            mydmp    all      custom 500 dump.lammpstrj id type x y z # Write atom coordinates to file
```

Visualization

```
# Uncomment to print atom coordinate to file
dump             mydmp    all      custom 500 dump.lammpstrj id type x y z # Write atom coordinates to file
```

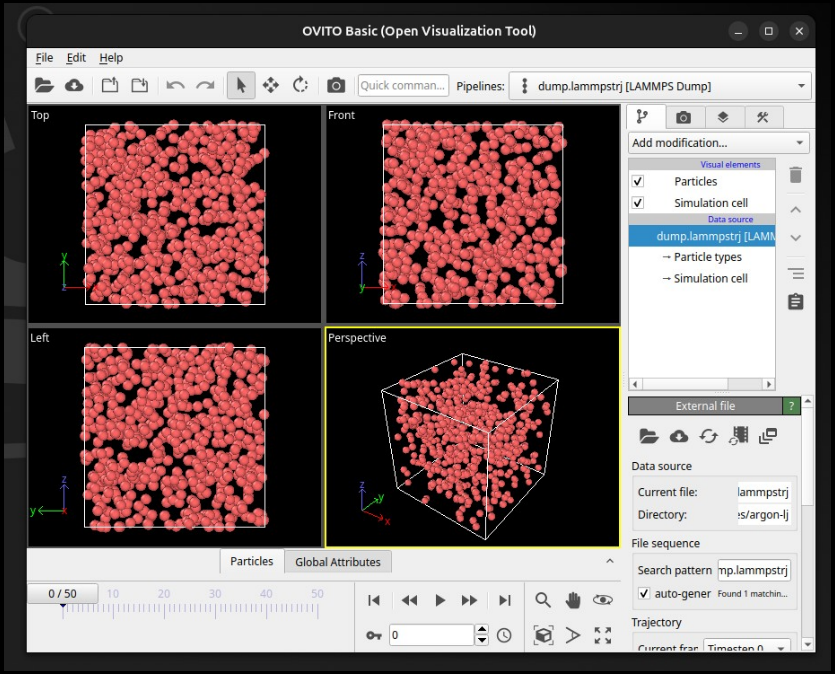
Can be opened with Ovito



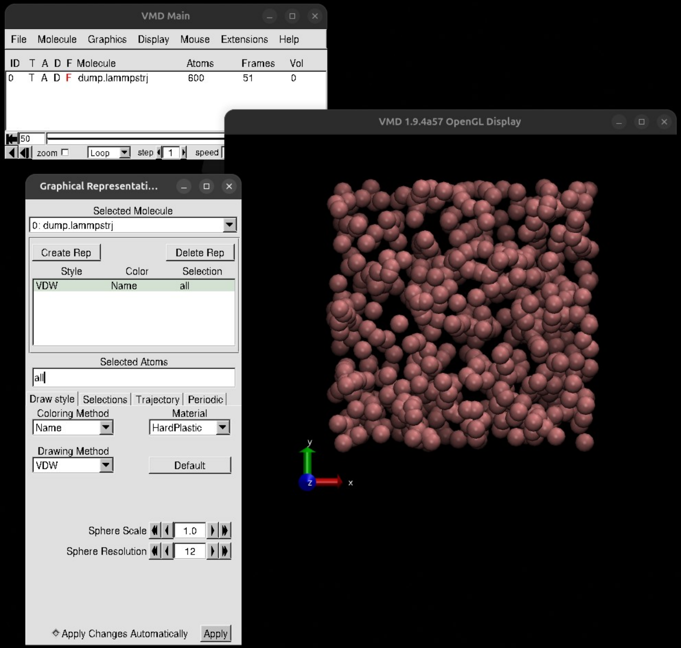
Visualization

```
# Uncomment to print atom coordinate to file
dump                mydmp    all      custom 500 dump.lammpstrj id type x y z # Write atom coordinates to file
```

Can be opened with Ovito



or VMD

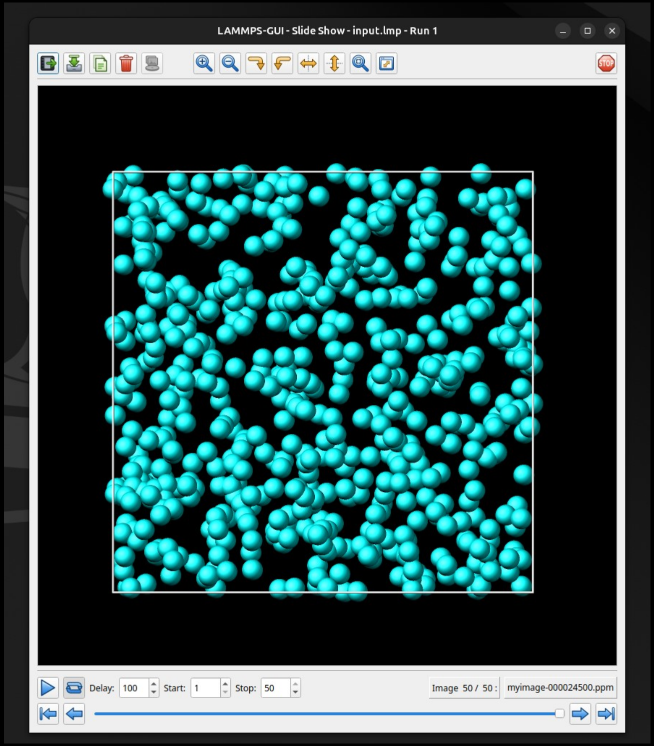


Visualization

```
# Uncomment to generate images
dump            viz      all      image 500 myimage-*.ppm type type size 800 800 zoom 1.452 shiny 0.5 fsaa yes view 0 0 box yes 0.005 axes no 0.0 0.0 center s 0.483725 0.510373 0.510373
dump_modify     viz pad 9 boxcolor white backcolor black adiam 1 2 acolor 1 cyan
```

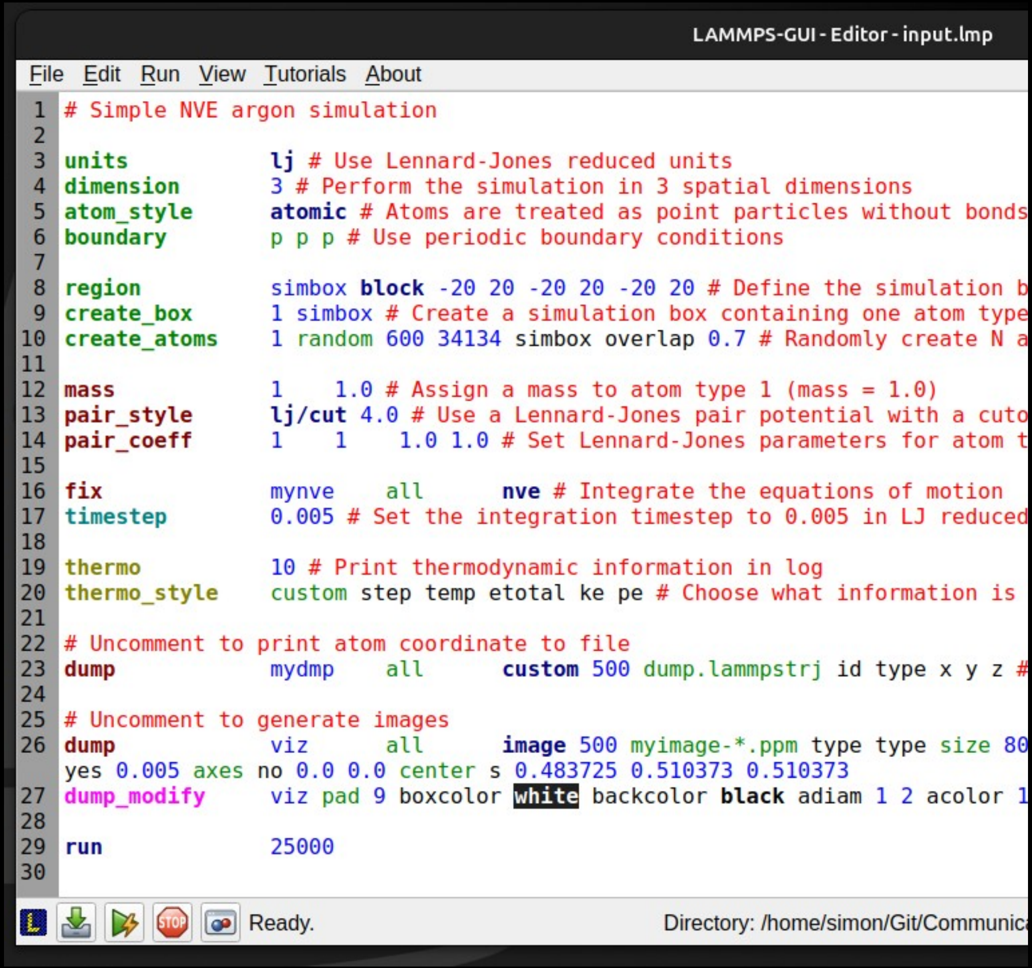
Visualization

```
# Uncomment to generate images
dump            viz       all       image 500 myimage-*.ppm type type size 800 800 zoom 1.452 shiny 0.5 fsaa yes view 0 0 box
yes 0.005 axes no 0.0 0.0 center s 0.483725 0.510373 0.510373
dump_modify     viz pad 9 boxcolor white backcolor black adiam 1 2 acolor 1 cyan
```



Select view → Slide show window

Play with the input



Play with the input

LAMMPS-GUI - Editor - input.lmp

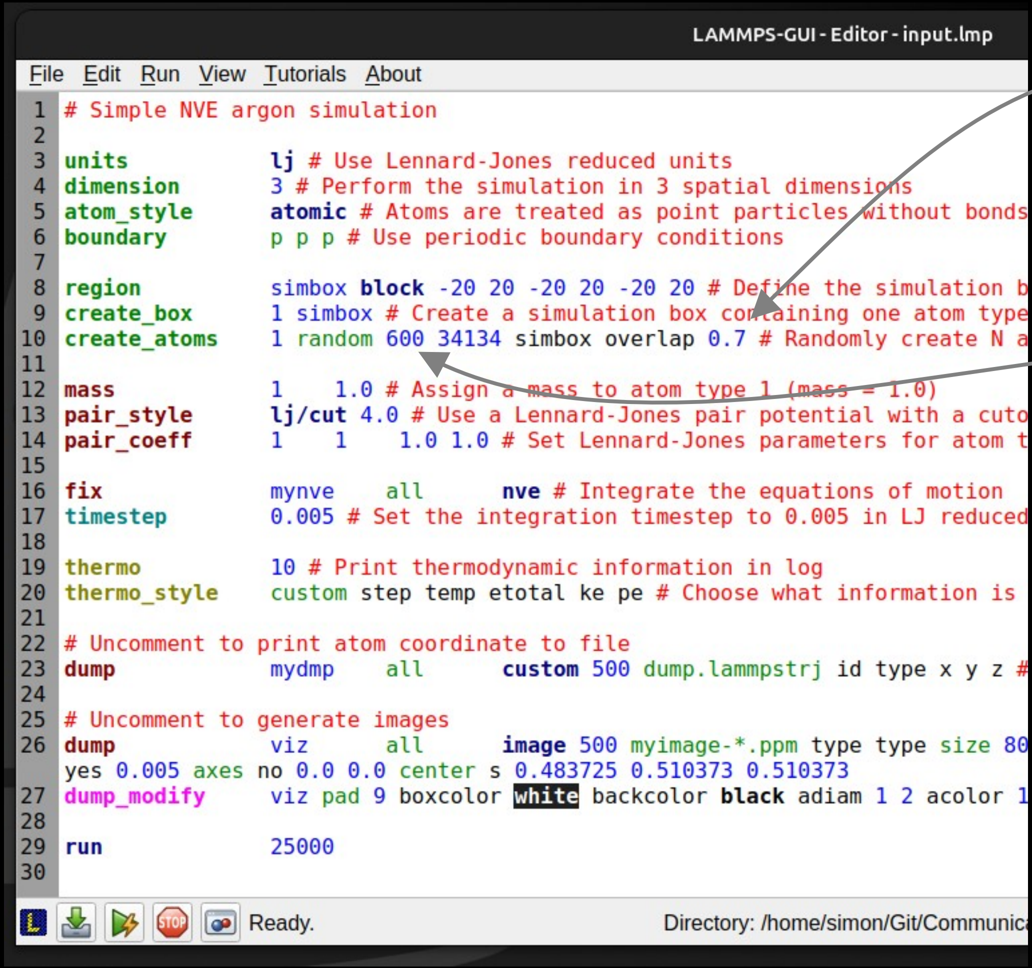
File Edit Run View Tutorials About

```
# Simple NVE argon simulation

units           lj # Use Lennard-Jones reduced units
dimension       3 # Perform the simulation in 3 spatial dimensions
atom_style      atomic # Atoms are treated as point particles without bonds
boundary        p p p # Use periodic boundary conditions

region          simbox block -20 20 -20 20 -20 20 # Define the simulation b
create_box      1 simbox # Create a simulation box containing one atom type
create_atoms    1 random 600 34134 simbox overlap 0.7 # Randomly create N a

mass            1    1.0 # Assign a mass to atom type 1 (mass = 1.0)
pair_style      lj/cut 4.0 # Use a Lennard-Jones pair potential with a cuto
pair_coeff      1    1    1.0 1.0 # Set Lennard-Jones parameters for atom t

fix             mynve    all      nve # Integrate the equations of motion
timestep        0.005 # Set the integration timestep to 0.005 in LJ reduced

thermo          10 # Print thermodynamic information in log
thermo_style    custom step temp etotal ke pe # Choose what information is

# Uncomment to print atom coordinate to file
dump            mydmp    all      custom 500 dump.lammpstrj id type x y z #

# Uncomment to generate images
dump            viz      all      image 500 myimage-*.ppm type type size 80
yes 0.005 axes no 0.0 0.0 center s 0.483725 0.510373 0.510373
dump_modify     viz pad 9 boxcolor white backcolor black adiam 1 2 acolor 1

run             25000
```

Ready. Directory: /home/simon/Git/Communica

Reduce / Increase overlap

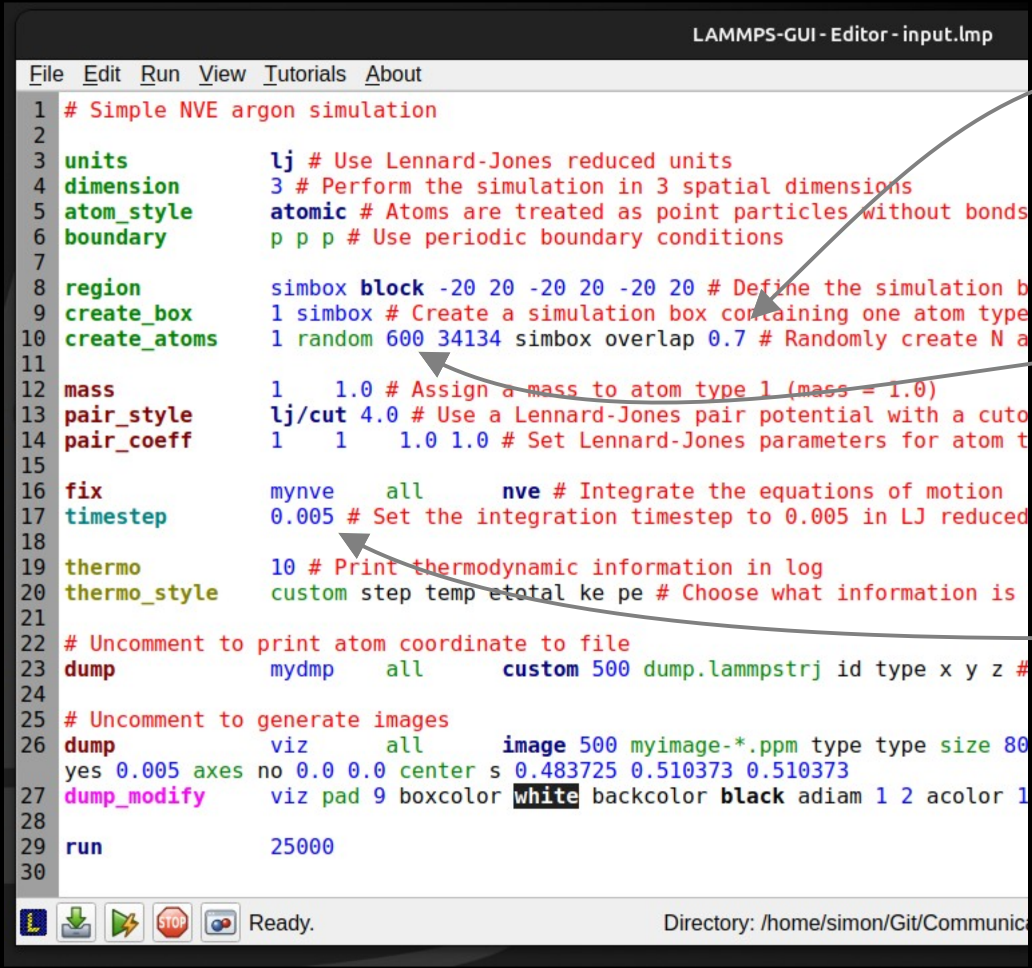
Play with the input



Reduce / Increase overlap

Change particle numbers

Play with the input

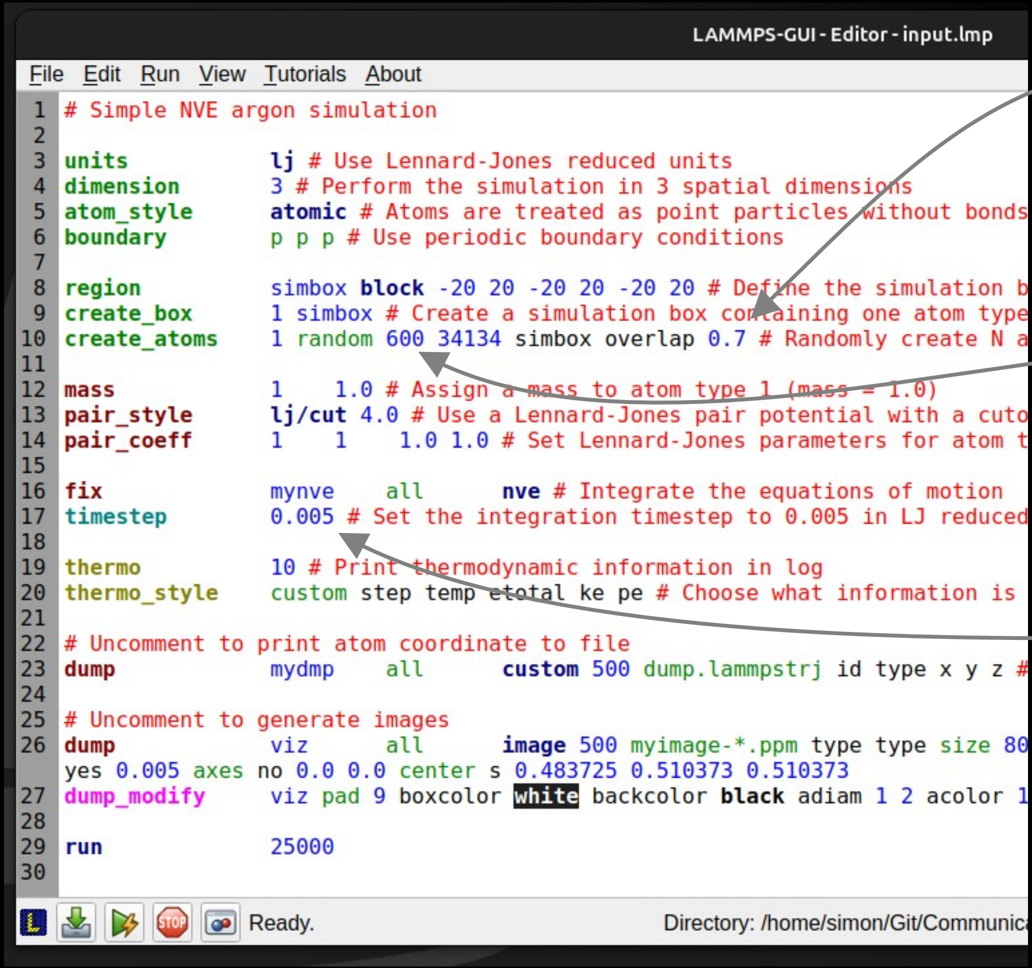


Reduce / Increase overlap

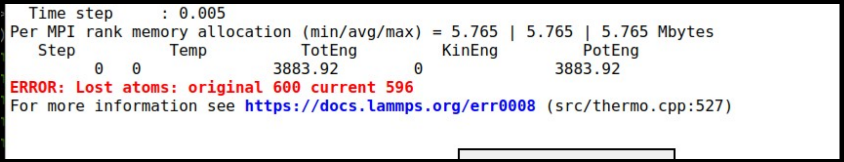
Change particle numbers

Use large timestep

Play with the input



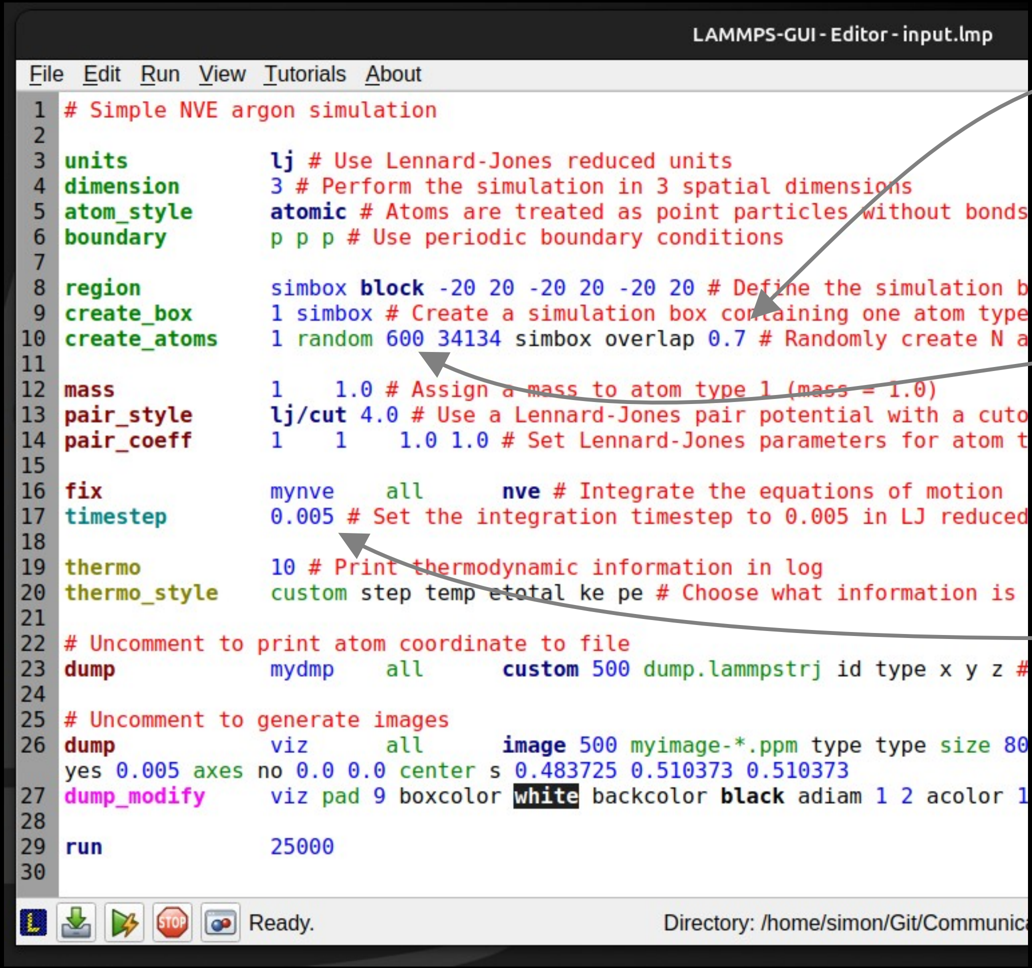
Reduce / Increase overlap



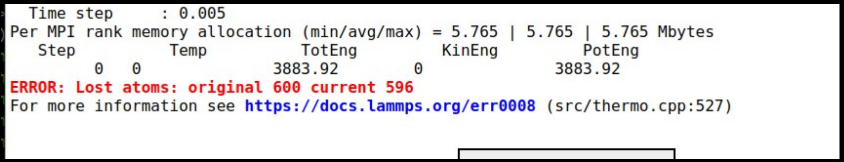
Change particle numbers

Use large timestep

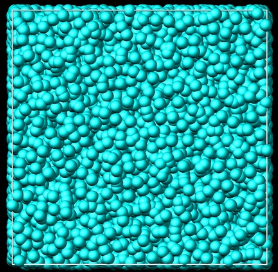
Play with the input



Reduce / Increase overlap



Change particle numbers



Use large timestep

Play with the input

LAMMPS-GUI - Editor - input.lmp

File Edit Run View Tutorials About

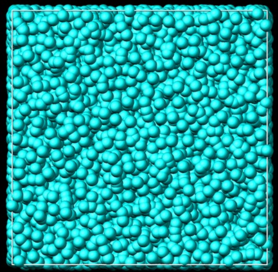
```
# Simple NVE argon simulation

units           lj # Use Lennard-Jones reduced units
dimension       3 # Perform the simulation in 3 spatial dimensions
atom_style      atomic # Atoms are treated as point particles without bonds
boundary        p p p # Use periodic boundary conditions

region          simbox block -20 20 -20 20 -20 20 # Define the simulation b
create_box      1 simbox # Create a simulation box containing one atom type
create_atoms    1 random 600 34134 simbox overlap 0.7 # Randomly create N a

mass            1    1.0 # Assign a mass to atom type 1 (mass = 1.0)
pair_style      lj/cut 4.0 # Use a Lennard-Jones pair potential with a cuto
pair_coeff      1    1    1.0 1.0 # Set Lennard-Jones parameters for atom t

fix             mynve    all      nve # Integrate the equations of motion
timestep        0.005 # Set the integration timestep to 0.005 in LJ reduced

thermo          10 # Print thermodynamic information in log
thermo_style    custom step temp etotal ke pe # Choose what information is

# Uncomment to print atom coordinate to file
dump            mydmp    all      custom 500 dump.lammpstrj id type x y z #

# Uncomment to generate images
dump            viz      all      image 500 myimage-*.ppm type type size 80
yes 0.005 axes no 0.0 0.0 center s 0.483725 0.510373 0.510373
dump_modify     viz pad 9 boxcolor white backcolor black adiam 1 2 acolor 1

run             25000
```

Ready.   Directory: /home/simon/Git/Communica

Reduce / Increase overlap

```
 Time step      : 0.005
Per MPI rank memory allocation (min/avg/max) = 5.765 | 5.765 | 5.765 Mbytes
   Step          Temp          TotEng          KinEng          PotEng
         0   0          3883.92        0                3883.92
ERROR: Lost atoms: original 600 current 596
For more information see https://docs.lammps.org/err0008 (src/thermo.cpp:527)
```

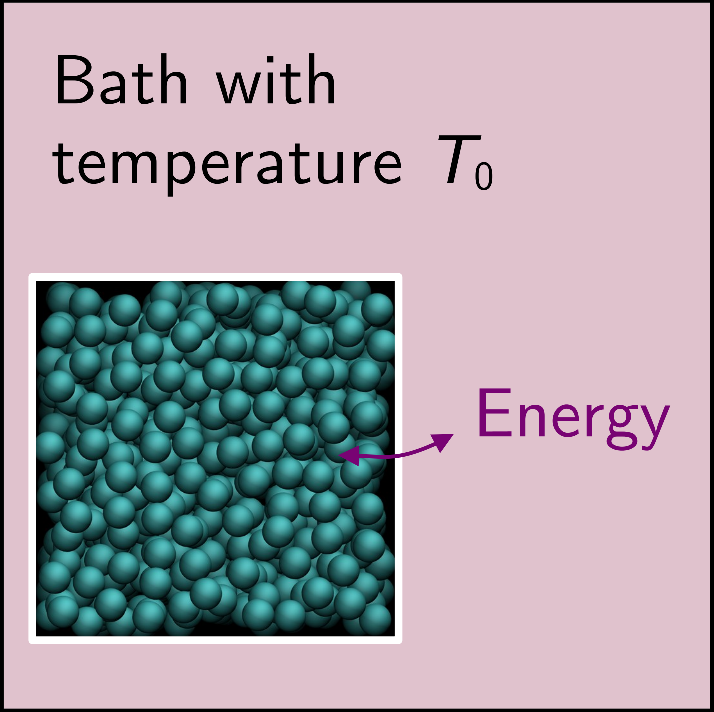
Change particle numbers



Use large timestep

```
 Time step      : 0.05
Per MPI rank memory allocation (min/avg/max) = 5.765 | 5.765 | 5.765 Mbytes
   Step          Temp          TotEng          KinEng          PotEng
         0   0          0.67828077      0                0.67828077
ERROR: Lost atoms: original 600 current 598
For more information see https://docs.lammps.org/err0008 (src/thermo.cpp:527)
```

Use a thermostat (NVT ensemble)

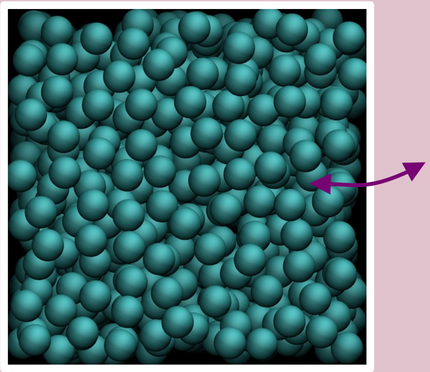


$$\mathbf{v}_i \rightarrow \lambda \mathbf{v}_i$$

$$\lambda = \sqrt{1 + \frac{\Delta t}{\tau}\left(\frac{T_0}{T} - 1\right)}$$

Use a thermostat (NVT ensemble)

Bath with temperature T_0



Energy

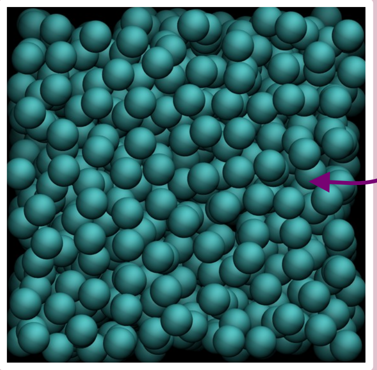
```
fix             mynve   all     nve # Integrate the equation
timestep        0.005 # Set the integration timestep to 0.005
fix             myber   all     temp/berendsen 1 1 10 # Ther

thermo          10 # Print thermodynamic information in log
thermo_style    custom step temp etotal ke pe # Choose what in
```

$$\mathbf{v}_i \to \lambda \mathbf{v}_i$$

$$\lambda = \sqrt{1 + \frac{\Delta t}{\tau}\left(\frac{T_0}{T} - 1\right)}$$

Use a thermostat (NVT ensemble)

Bath with
temperature T_0



Energy

```
fix             mynve    all      nve # Integrate the equation
timestep        0.005 # Set the integration timestep to 0.005
fix             myber    all      temp/berendsen 1 1 10 # Ther

thermo          10 # Print thermodynamic information in log
thermo_style    custom step temp etotal ke pe # Choose what in
```

Initial temperature

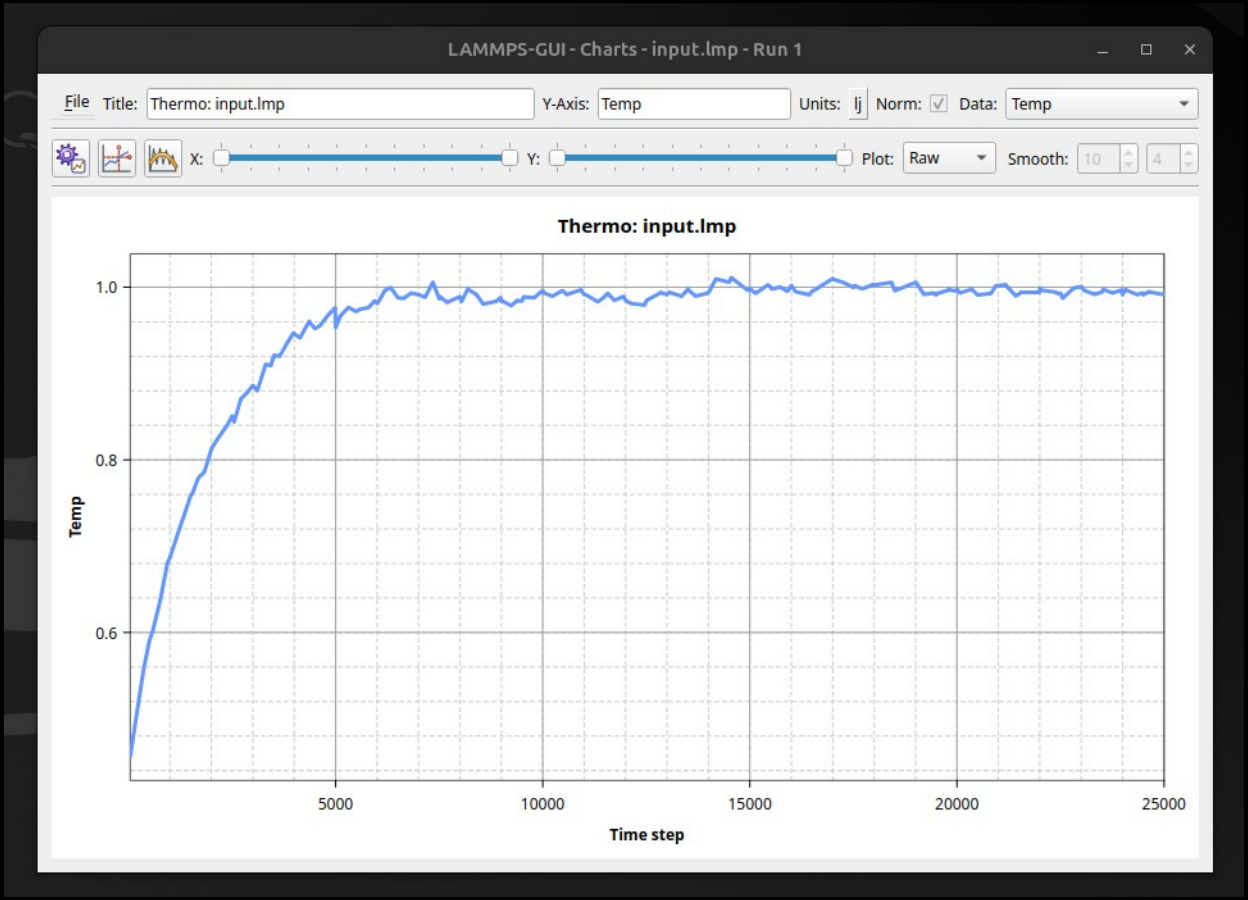
Final temperature

τ

$$\mathbf{v}_i \to \lambda \mathbf{v}_i$$

$$\lambda = \sqrt{1 + \frac{\Delta t}{\tau}\left(\frac{T_0}{T} - 1\right)}$$

Use a thermostat (NVT ensemble)



Use a barostat (NPT ensemble)

Use a barostat (NPT ensemble)

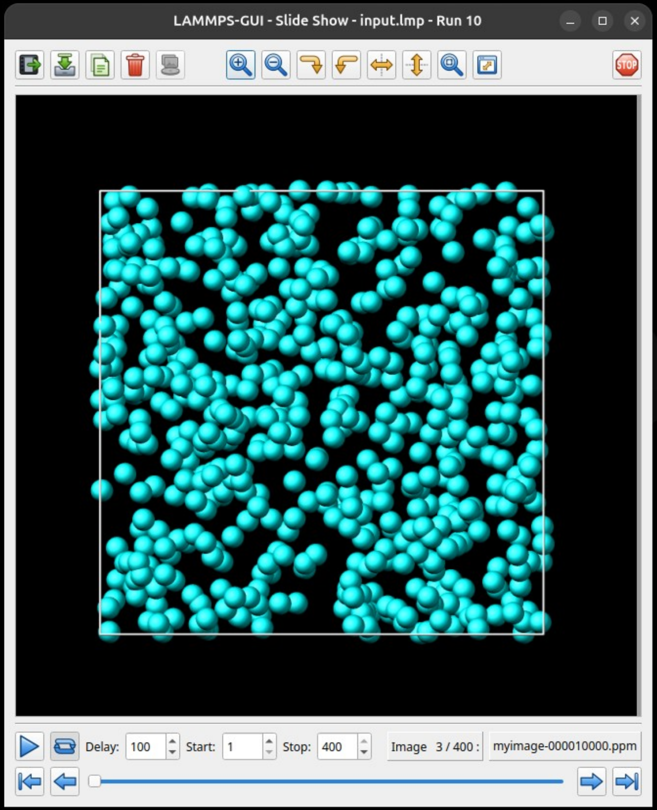
```
fix             mynve    all      nve # Integrate the equations of motion
timestep        0.005 # Set the integration timestep to 0.005 in LJ reduced time units
fix             mytber    all      temp/berendsen 1 1 10 # Thermostat at temperature 1 with time constant 10
fix             mypber   all      press/berendsen iso 1.0 1.0 100 # Barostat at pressure 1 with time constant 100

thermo          100 # Print thermodynamic information in log
thermo_style    custom step temp etotal ke pe density # Choose what information is printed

# Uncomment to print atom coordinate to file
dump            mydmp    all      custom 5000 dump.lammpstrj id type x y z # Write atom coordinates to file

# Uncomment to generate images
dump            viz      all      image 5000 myimage-*.ppm type type size 800 800 zoom 1.452 shiny 0.5 fsaa yes view 0 0 box yes 0.005 axes no 0.0 0.0 center s 0.483725 0.510373 0.510373
dump_modify     viz pad 9 boxcolor white backcolor black adiam 1 2 acolor 1 cyan

run             2000000
```

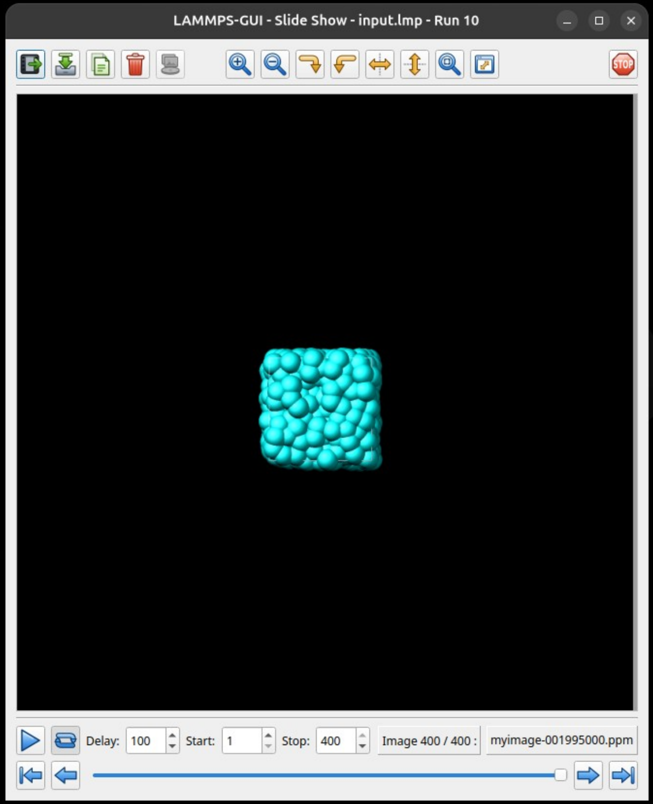
Use a barostat (NPT ensemble)

```
fix             mynve    all      nve # Integrate the equations of motion
timestep        0.005 # Set the integration timestep to 0.005 in LJ reduced time units
fix             mytber    all      temp/berendsen 1 1 10 # Thermostat at temperature 1 with time constant 10
fix             mypber   all      press/berendsen iso 1.0 1.0 100 # Barostat at pressure 1 with time constant 100

thermo          100 # Print thermodynamic information in log
thermo_style    custom step temp etotal ke pe density # Choose what information is printed

# Uncomment to print atom coordinate to file
dump            mydmp    all      custom 5000 dump.lammpstrj id type x y z # Write atom coordinates to file

# Uncomment to generate images
dump            viz      all      image 5000 myimage-*.ppm type type size 800 800 zoom 1.452 shiny 0.5 fsaa yes
view 0 0 box yes 0.005 axes no 0.0 0.0 center s 0.483725 0.510373 0.510373
dump_modify     viz pad 9 boxcolor white backcolor black adiam 1 2 acolor 1 cyan

run             2000000
```

Increase simulation duration

Use a barostat (NPT ensemble)





Measuring RDF

$$g(r) = \frac{1}{4\pi r^2 \rho N} \left\langle \sum_{i=1}^{N} \sum_{j \neq i} \delta\left(r - r_{ij}\right) \right\rangle$$

Measuring RDF

$$g(r) = \frac{1}{4\pi r^2 \rho N} \left\langle \sum_{i=1}^{N} \sum_{j\neq i} \delta\left(r - r_{ij}\right) \right\rangle$$

```
run             25000 # First run to equilibrate the system

compute         myRDF    all      rdf 50
fix             myAT        all      ave/time 10 2500 25000 c_myRDF[*] file rdf.dat mode vector

run             25000 # Second run to produce
```

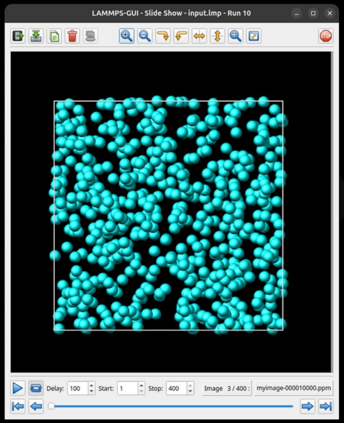
Available from GitHub

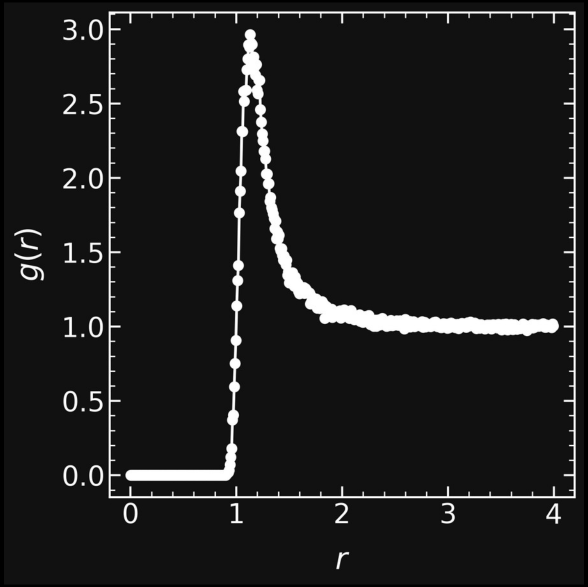
github.com/simongravelle/DIADEM

Measuring RDF

$$g(r) = \frac{1}{4\pi r^2 \rho N} \left\langle \sum_{i=1}^{N} \sum_{j \neq i} \delta\left(r - r_{ij}\right) \right\rangle$$

```
run             25000 # First run to equilibrate the system

compute         myRDF    all      rdf 50
fix             myAT        all      ave/time 10 2500 25000 c_myRDF[*] file rdf.dat mode vector

run             25000 # Second run to produce
```

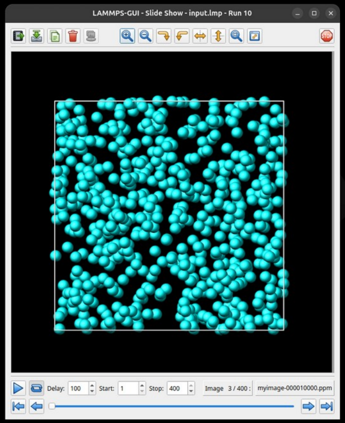


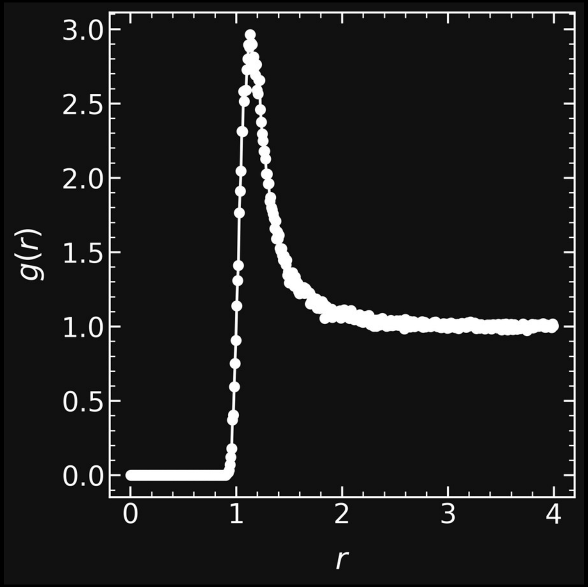


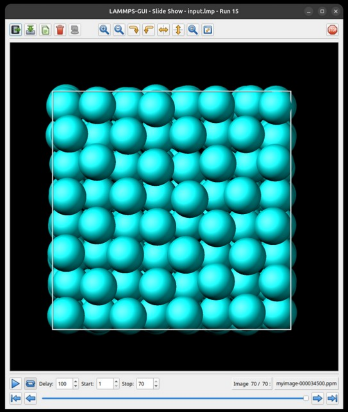
Measuring RDF

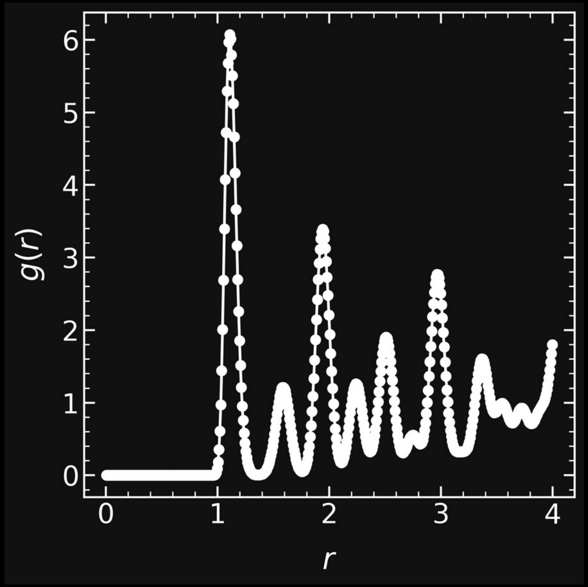
$$g(r) = \frac{1}{4\pi r^2 \rho N} \left\langle \sum_{i=1}^{N} \sum_{j \neq i} \delta\left(r - r_{ij}\right) \right\rangle$$

```
run             25000 # First run to equilibrate the system

compute         myRDF    all      rdf 50
fix             myAT        all      ave/time 10 2500 25000 c_myRDF[*] file rdf.dat mode vector

run             25000 # Second run to produce
```

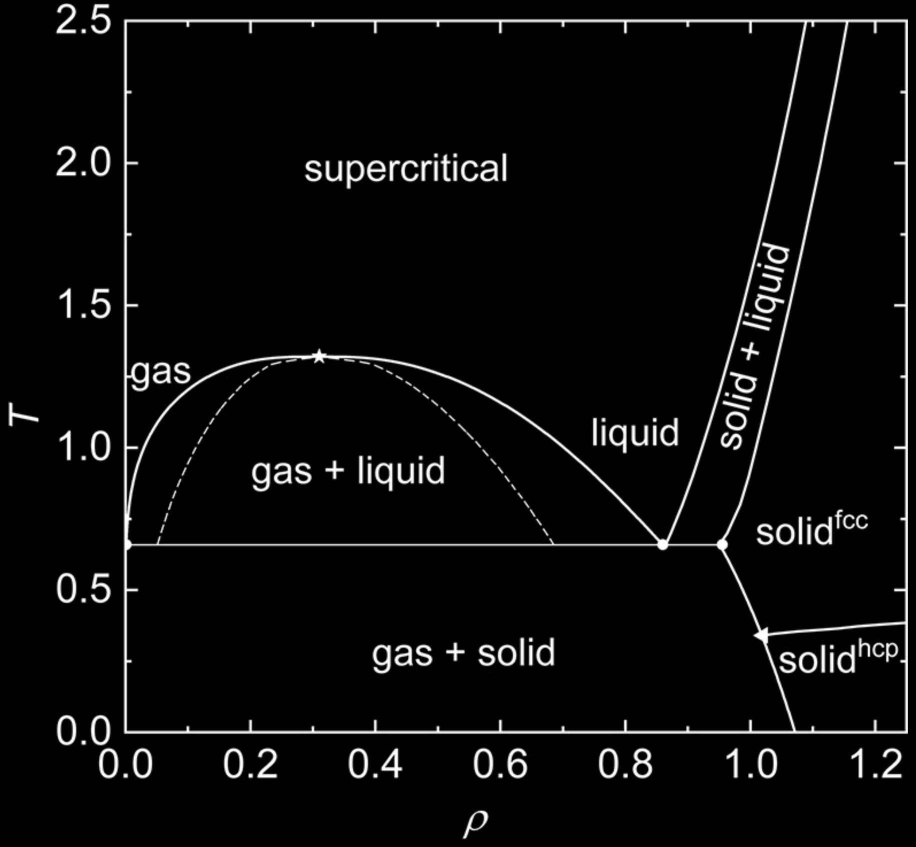




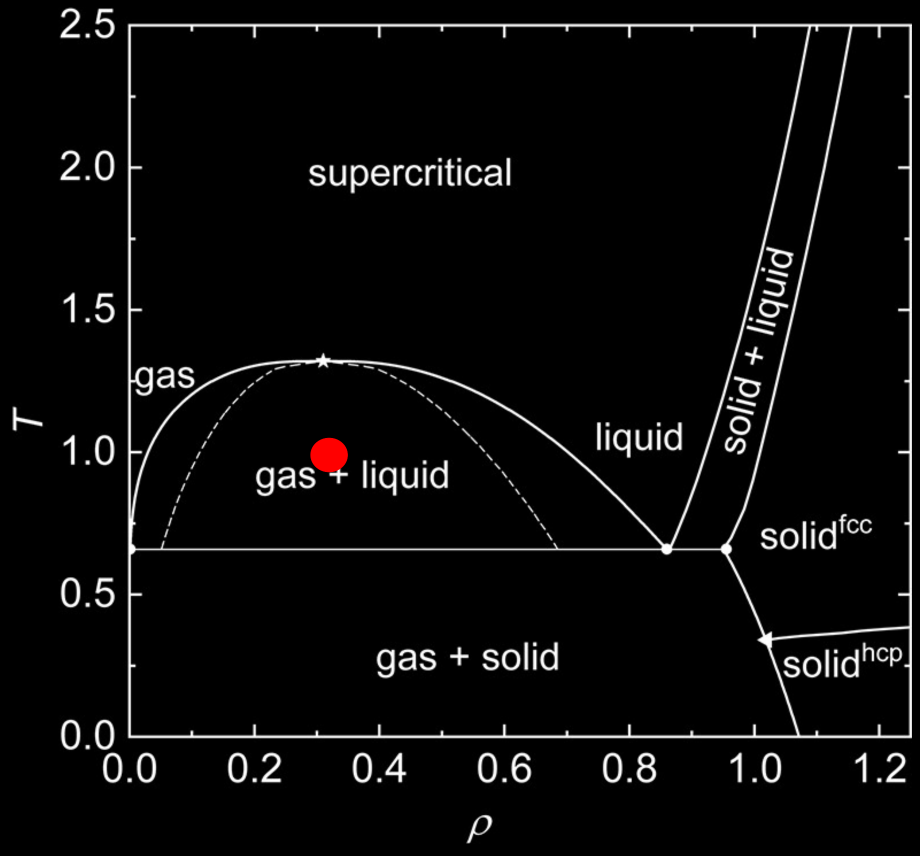




Explore the phase diagram of LJ



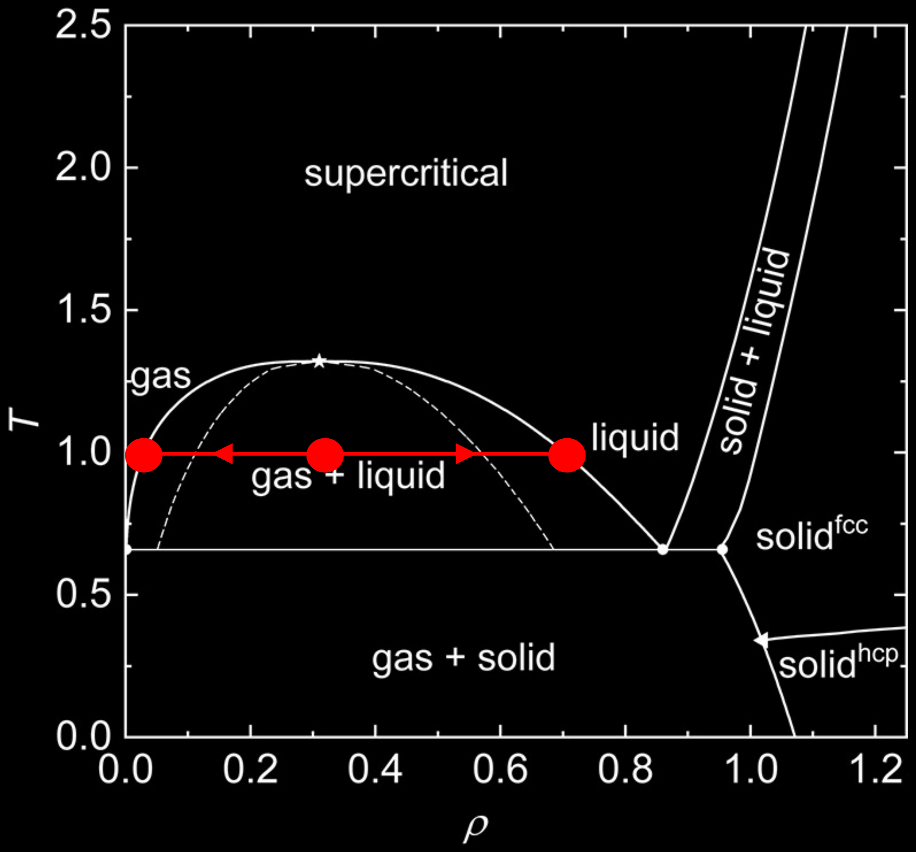
Explore the phase diagram of LJ



Exercise:

Generate a multiphase system

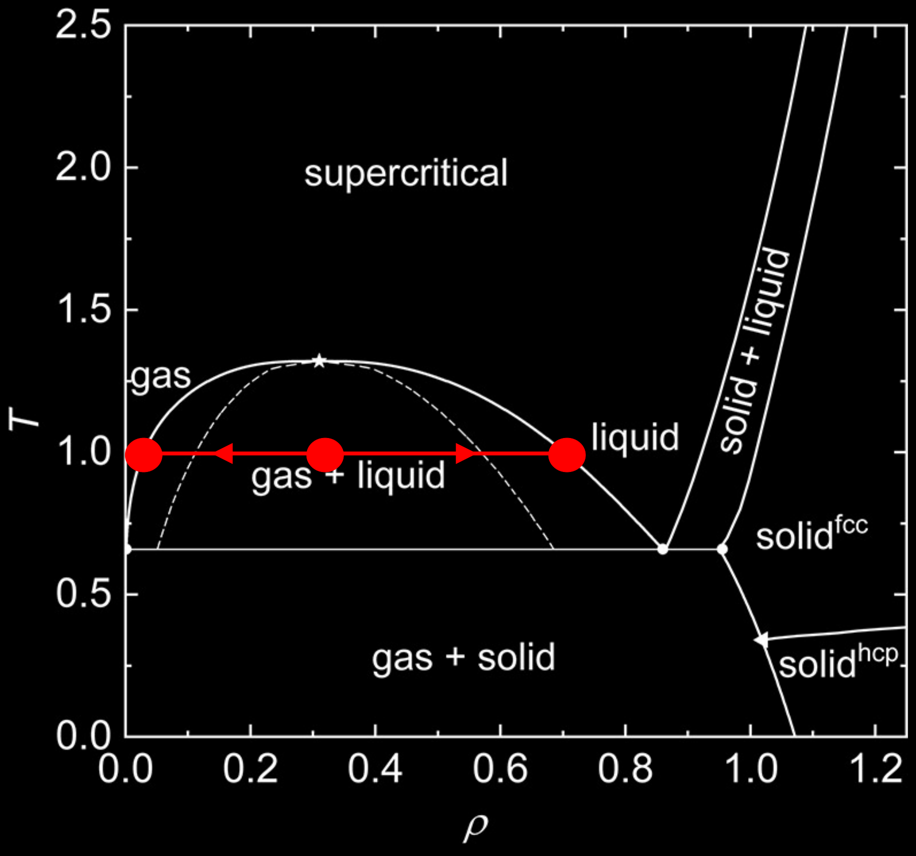
Explore the phase diagram of LJ



Exercise:

Generate a multiphase system

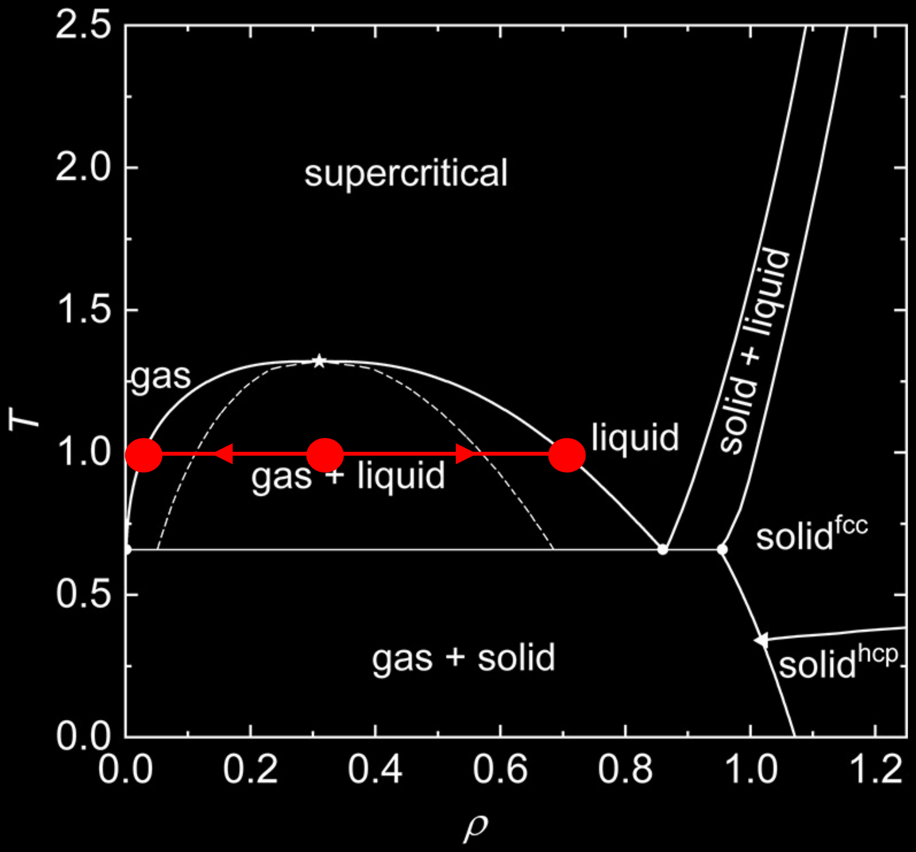
Explore the phase diagram of LJ



Exercise:

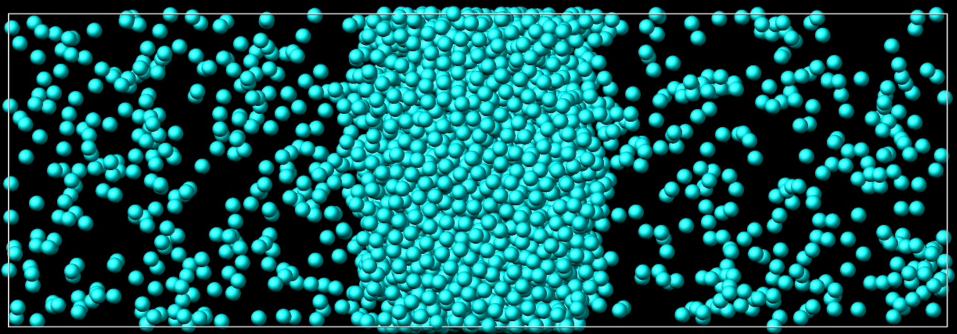
Generate a **slab** multiphase system

Explore the phase diagram of LJ

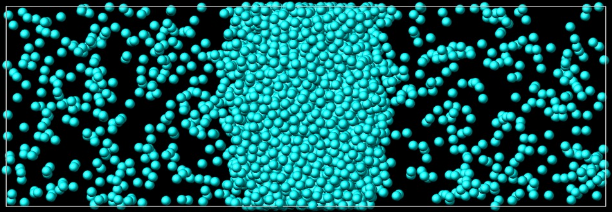


Exercise:

Generate a **slab** multiphase system



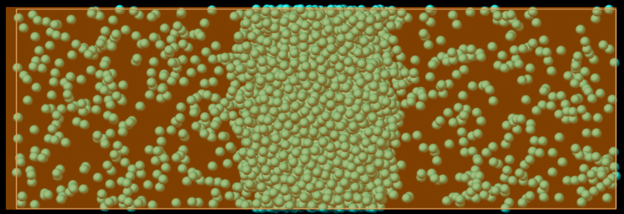
Generate a **slab** multiphase system



1) Create 3 regions

```
region          simbox block -10 10 -30 30 -10 10 # Defi
create_box      1 simbox # Create a simulation box conta

region          liquid block -10 10 -5 5 -10 10 # Define
create_atoms    1 random 2800 82234 liquid overlap 0.9 #

region          vapor block -10 10 -5 5 -10 10 side out
create_atoms    1 random 1400 34134 vapor overlap 0.9 #
```

Generate a **slab** multiphase system

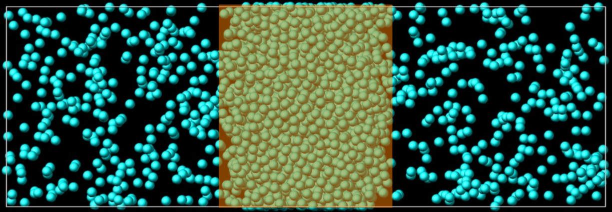


simbox

1) Create 3 regions

```
region          simbox block -10 10 -30 30 -10 10 # Defi
create_box      1 simbox # Create a simulation box conta

region          liquid block -10 10 -5 5 -10 10 # Define
create_atoms    1 random 2800 82234 liquid overlap 0.9 #

region          vapor block -10 10 -5 5 -10 10 side out
create_atoms    1 random 1400 34134 vapor overlap 0.9 #
```

Generate a **slab** multiphase system

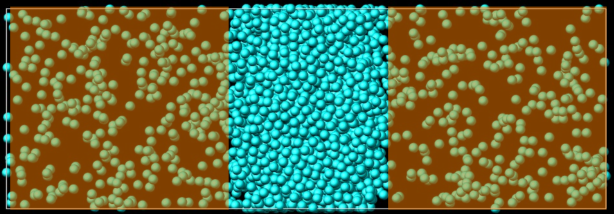


liquid

1) Create 3 regions

```
region          simbox block -10 10 -30 30 -10 10 # Defi
create_box      1 simbox # Create a simulation box conta

region          liquid block -10 10 -5 5 -10 10 # Define
create_atoms    1 random 2800 82234 liquid overlap 0.9 #

region          vapor block -10 10 -5 5 -10 10 side out
create_atoms    1 random 1400 34134 vapor overlap 0.9 #
```

Generate a **slab** multiphase system

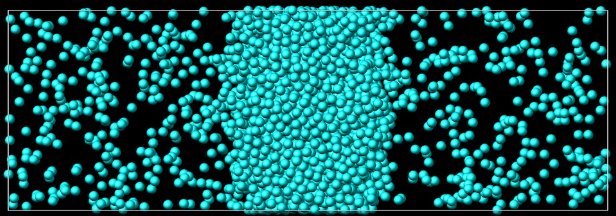


vapor

1) Create 3 regions

```
region            simbox block -10 10 -30 30 -10 10 # Defi
create_box        1 simbox # Create a simulation box conta

region            liquid block -10 10 -5 5 -10 10 # Define
create_atoms      1 random 2800 82234 liquid overlap 0.9 #

region            vapor block -10 10 -5 5 -10 10 side out
create_atoms      1 random 1400 34134 vapor overlap 0.9 #
```

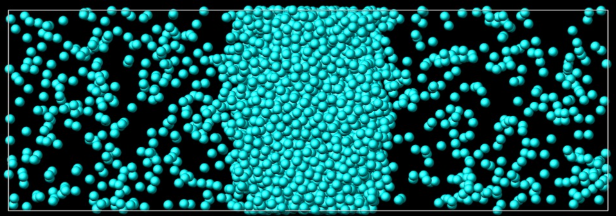
Generate a **slab** multiphase system



1) Create 3 regions

```
region          simbox block -10 10 -30 30 -10 10 # Defi
create_box      1 simbox # Create a simulation box conta

region          liquid block -10 10 -5 5 -10 10 # Define
create_atoms    1 random 2800 82234 liquid overlap 0.9 #

region          vapor block -10 10 -5 5 -10 10 side out
create_atoms    1 random 1400 34134 vapor overlap 0.9 #
```

Generate a **slab** multiphase system

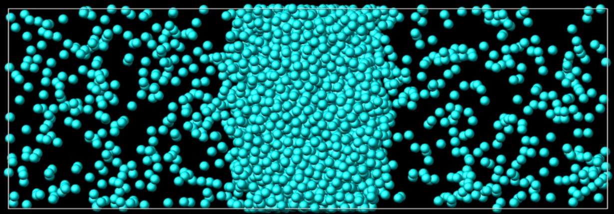


1) Create 3 regions

2) Minimize the system energy

```
mass            1    1.0 # Assign a mass to atom type 1 (mass =
pair_style      lj/cut 4.0 # Use a Lennard-Jones pair potential
pair_coeff      1    1    1.0 1.0 # Set Lennard-Jones parameter

minimize        1.0e-4 1.0e-6 100 1000 # reduce the initial ene

fix             mynve    all       nve # Integrate the equations
timestep        0.005 # Set the integration timestep to 0.005 i
fix             myber    all       temp/berendsen 1 1 10 # Therm
```

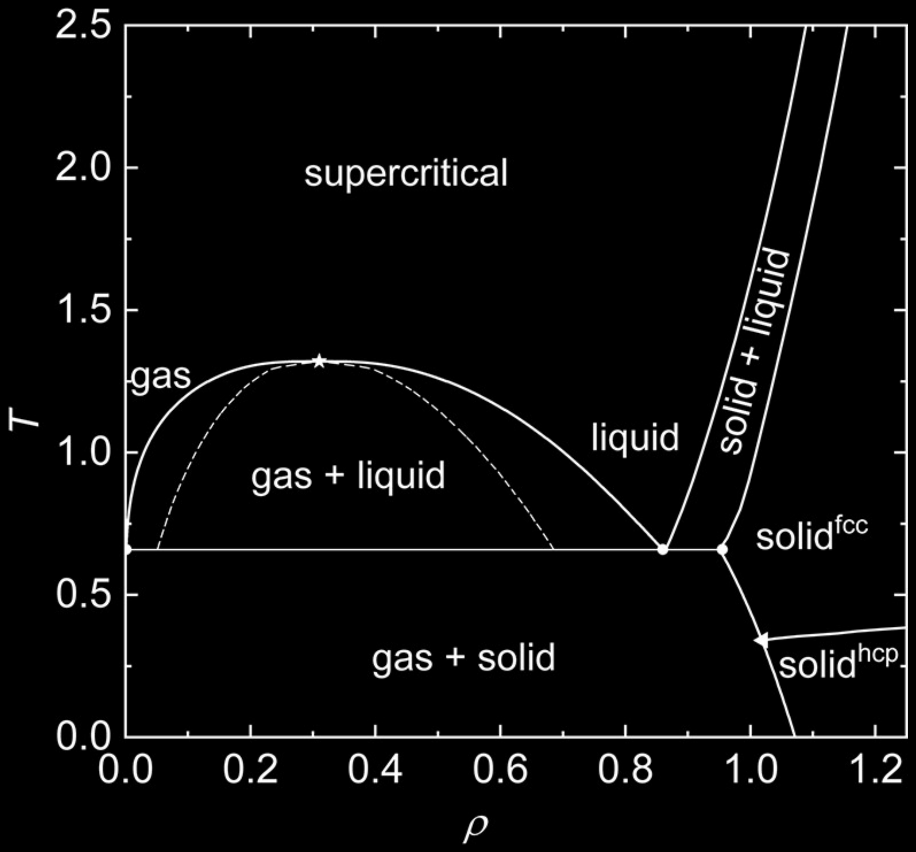
Generate a **slab** multiphase system



1) Create 3 regions
2) Minimize the system energy
3) Adjust visualization

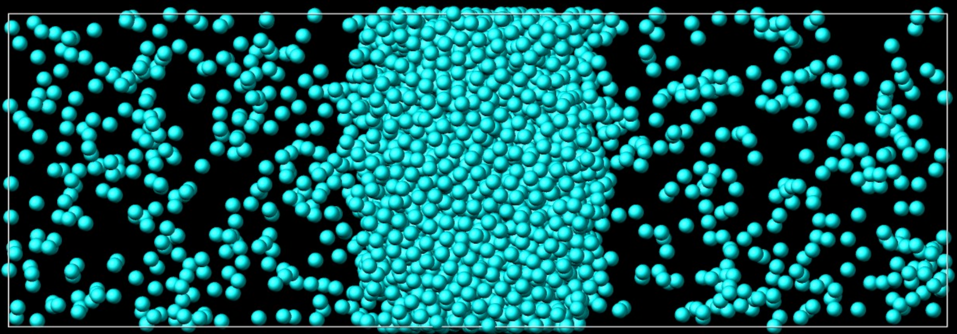
```
# Uncomment to generate images
dump            viz      all      image 500 myimage-*.ppm type type size 1600 800 zoom 3 shiny 0.5 fsaa yes view 0 0 box yes
0.005 axes no 0.0 0.0 center s 0.483725 0.510373 0.510373
dump_modify     viz pad 9 boxcolor white backcolor black adiam 1 1 acolor 1 cyan
```

Explore the phase diagram of LJ



Exercise:

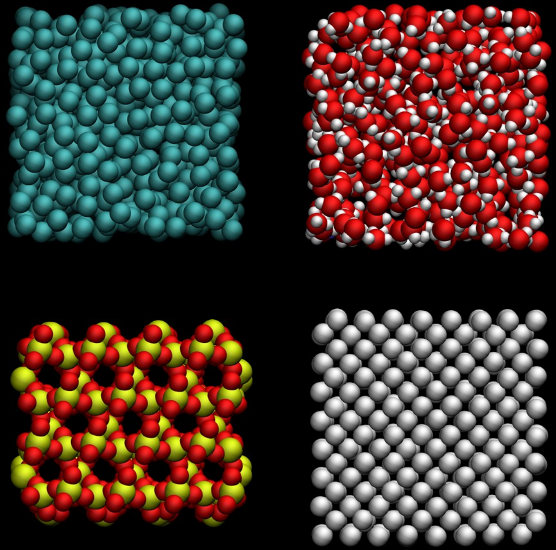
Generate a **slab** multiphase system



Explore the repository

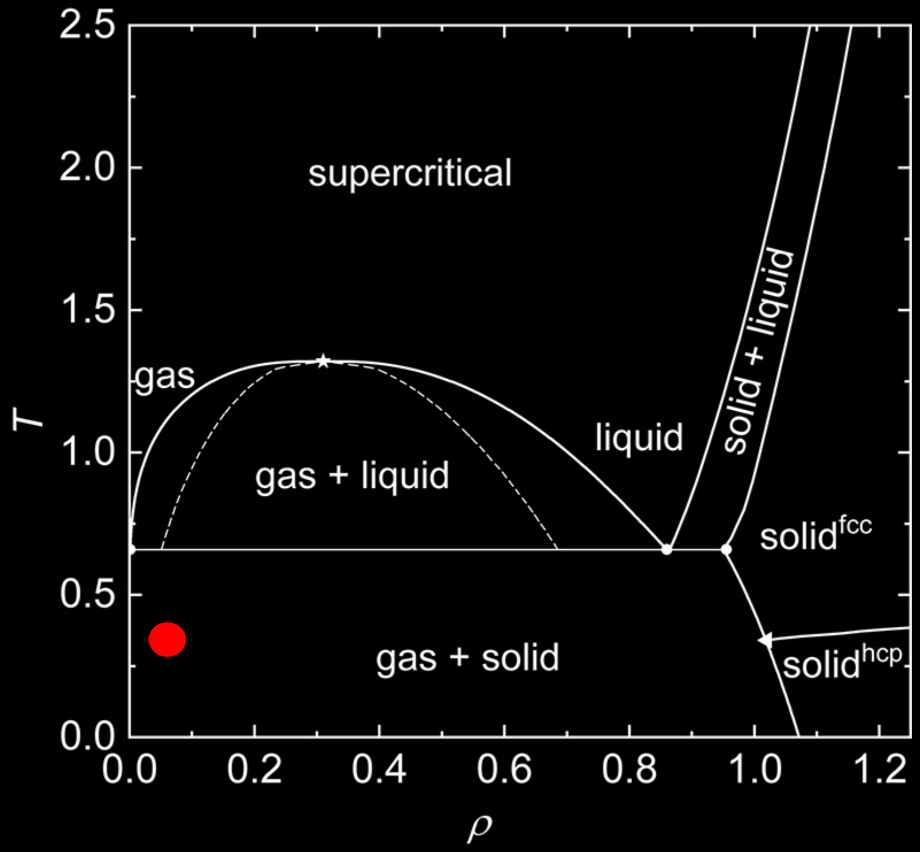
https://tinyurl.com/y9w53mrj (zip)

https://tinyurl.com/56mecmb3 (tar.gz)



https://github.com/simongravelle/DIADEM

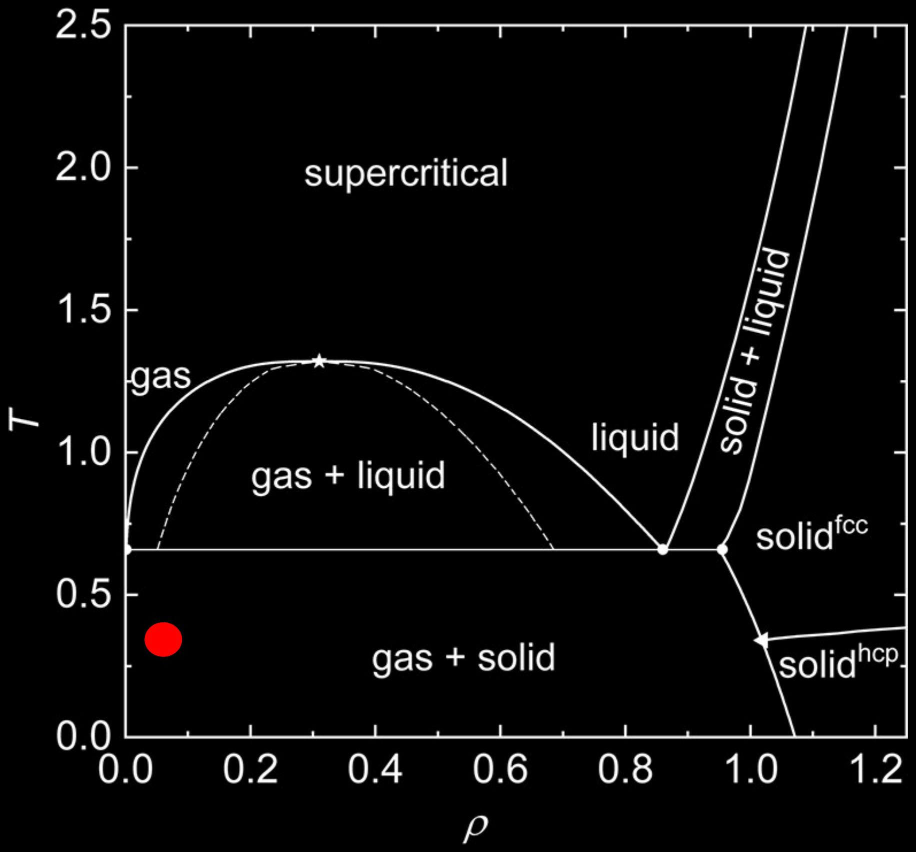
Explore the phase diagram of LJ



Exercise:

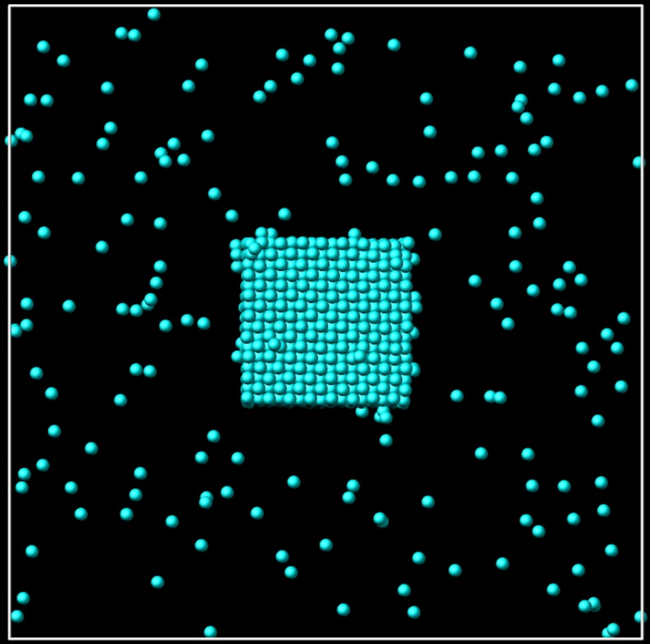
Generate a **solid/gas** multiphase system

Explore the phase diagram of LJ

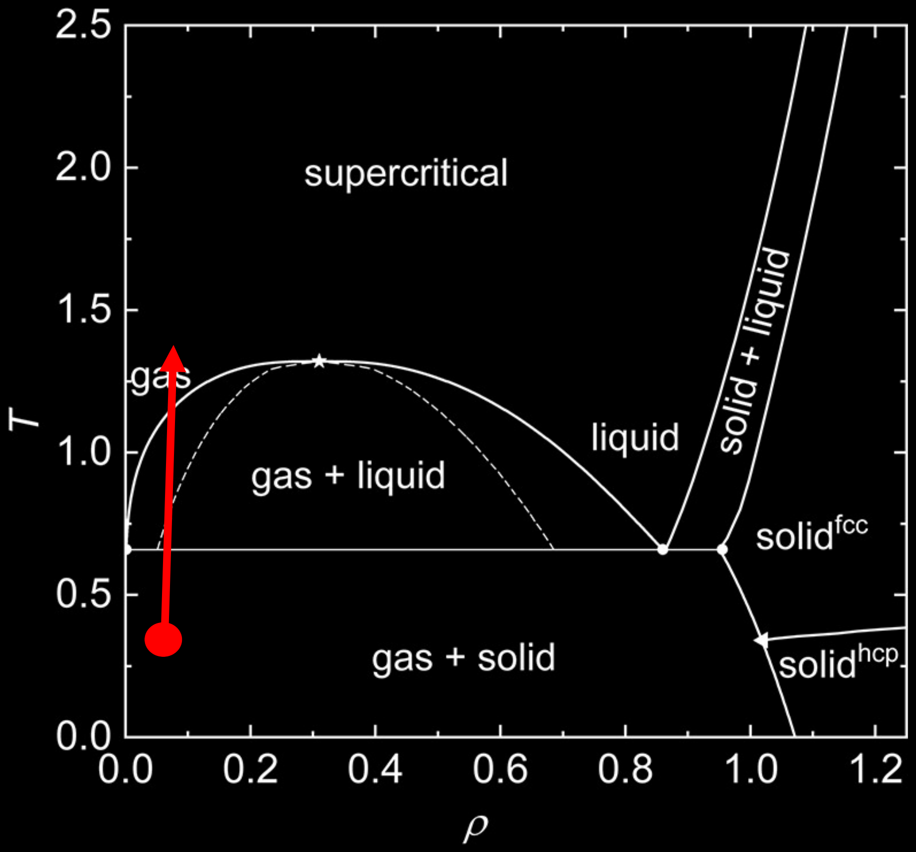


Exercise:

Generate a **solid/gas** multiphase system

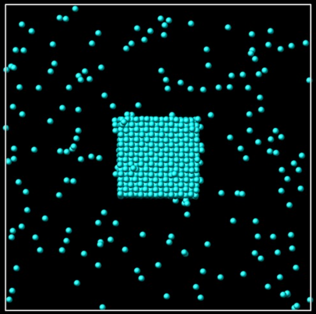


Explore the phase diagram of LJ

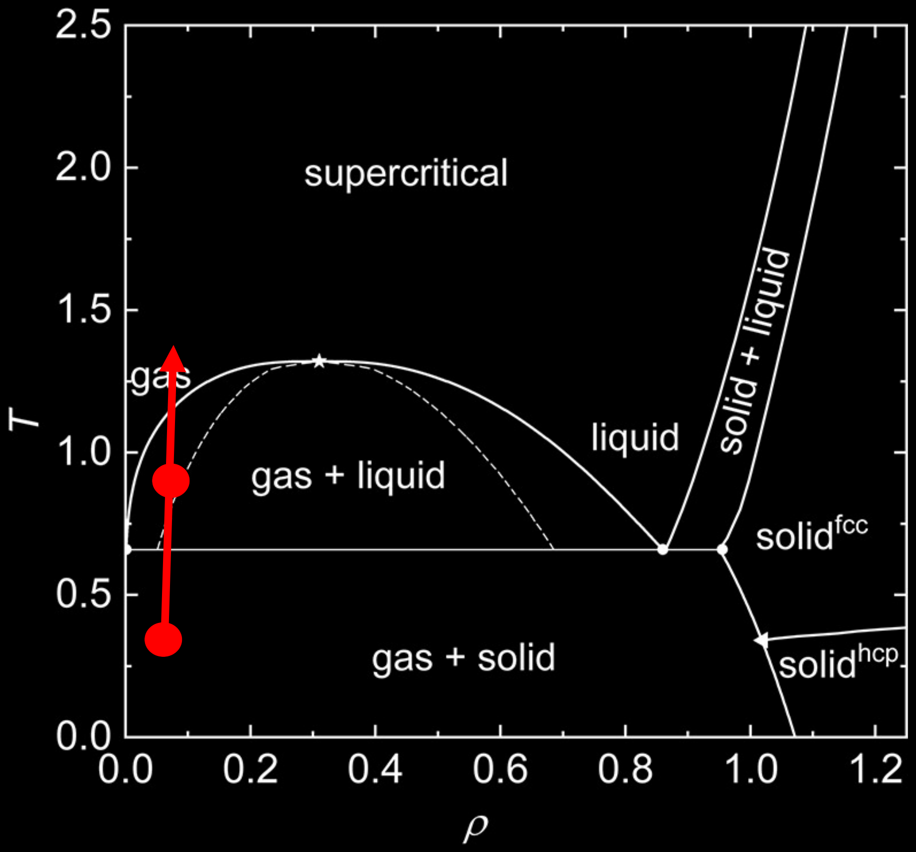


Exercise:

Generate a **solid/gas** multiphase system

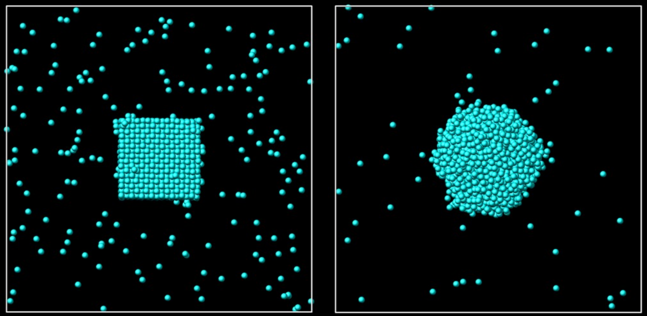


Explore the phase diagram of LJ

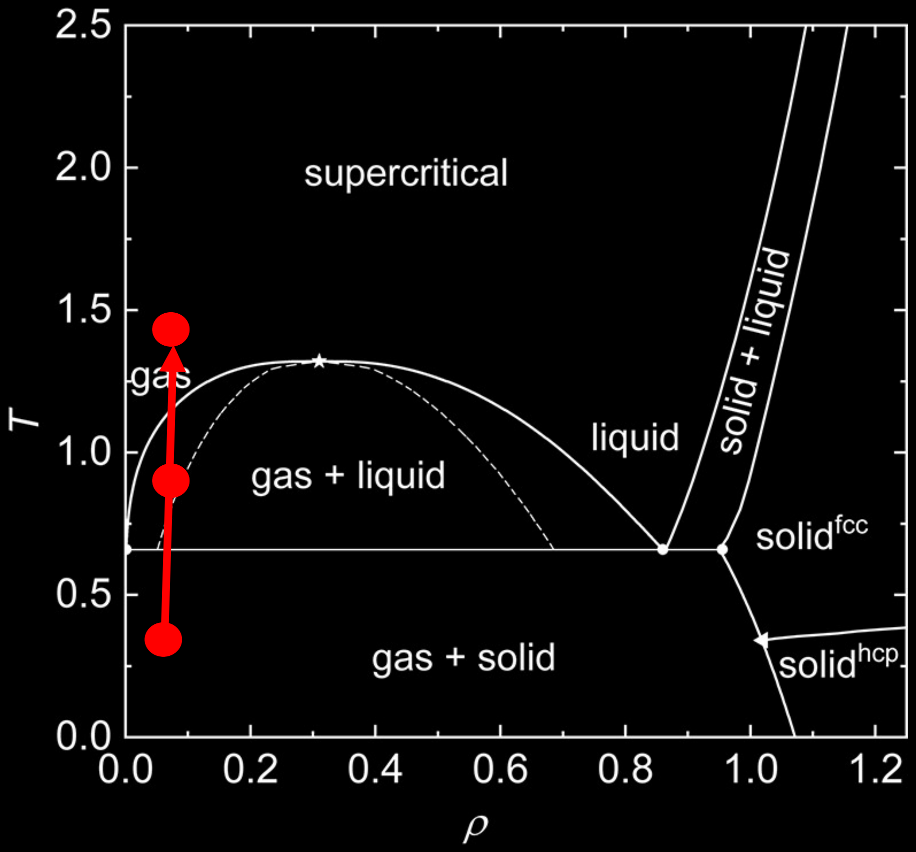


Exercise:

Generate a **solid/gas** multiphase system

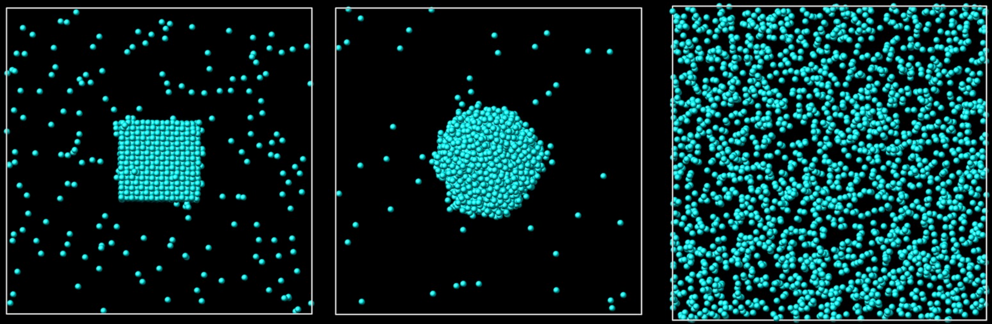


Explore the phase diagram of LJ



<u>Exercise</u>:

Generate a **solid**/**gas** multiphase system



Generate a **solid/gas** multiphase system

1) Create atoms on a lattice

```
lattice         fcc 1 origin 0.25 0.25 0.25 # Define a fcc lat
region          simbox block -15 15 -15 15 -15 15 # Define the
create_box      1 simbox # Create a simulation box containing

region          solid block -4 4 -4 4 -4 4 # Define a region
create_atoms    1 region solid # Create atoms on a lattice
```

Generate a **solid/gas** multiphase system

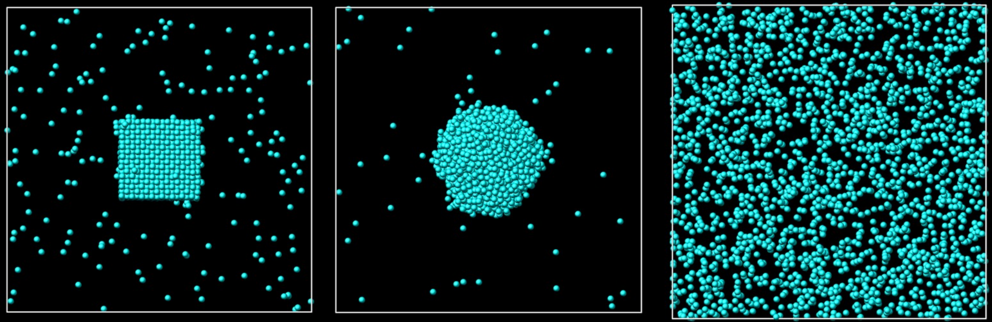
1) Create atoms on a lattice

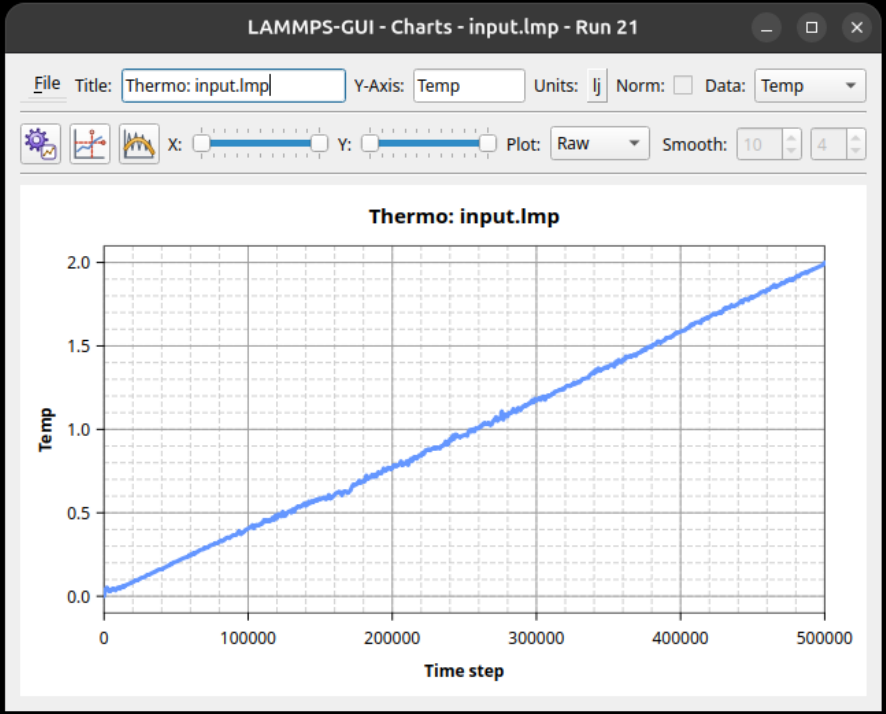
```
lattice         fcc 1 origin 0.25 0.25 0.25 # Define a fcc lat
region          simbox block -15 15 -15 15 -15 15 # Define the
create_box      1 simbox # Create a simulation box containing

region          solid block -4 4 -4 4 -4 4 # Define a region
create_atoms    1 region solid # Create atoms on a lattice
```

2) Use different init and final temperatures

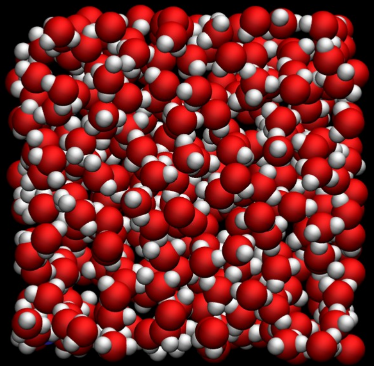
```
timestep        0.005 # Set the integration timestep to 0.005 in
fix             myber    all      temp/berendsen 0.02 2.0 10 # T
```

Generate a **solid/gas** multiphase system

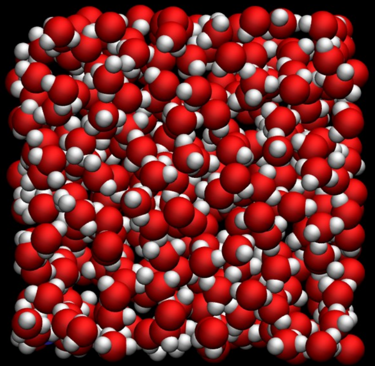


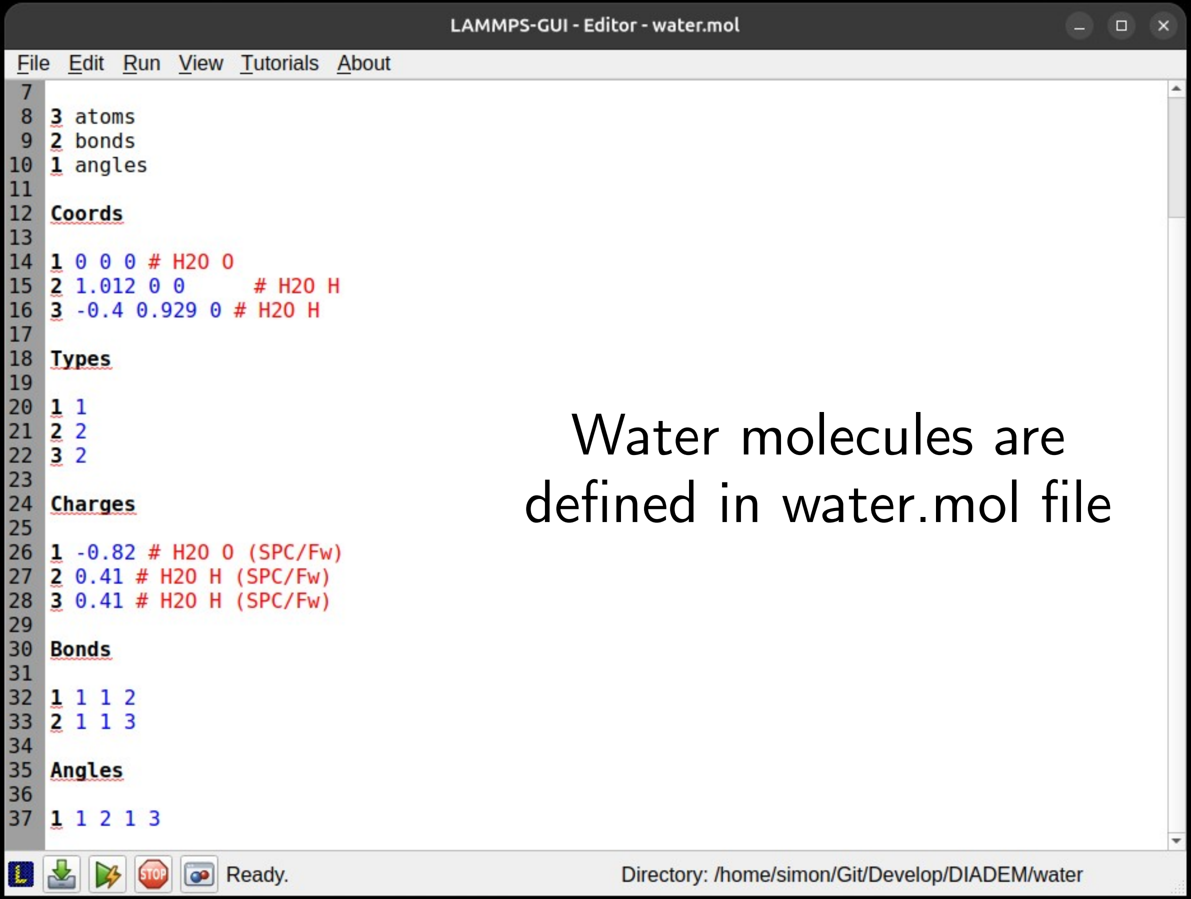


Simulating water



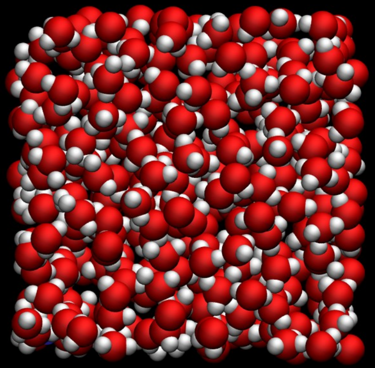
Simulating water





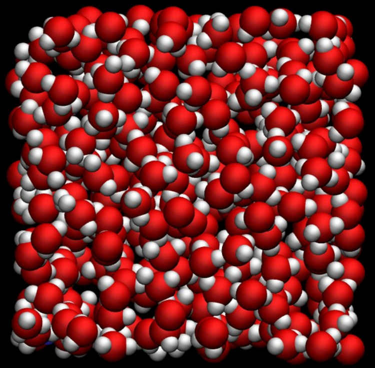
Water molecules are defined in water.mol file

Simulating water



```
units real
atom_style full
bond_style harmonic
angle_style harmonic
pair_style lj/cut/coul/long 10
kspace_style ewald 1e-5
```

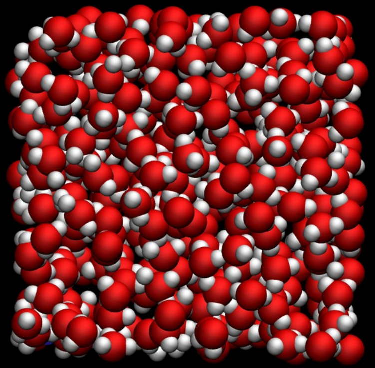
Simulating water



```
units real
atom_style full
bond_style harmonic
angle_style harmonic
pair_style lj/cut/coul/long 10
kspace_style ewald 1e-5
```

```
region box block -15 15 -15 15 -15 15
  create_box 2 box &
  bond/types 1 &
  angle/types 1 &
  extra/bond/per/atom 2 &
  extra/angle/per/atom 2 &
  extra/special/per/atom 2
```

Simulating water

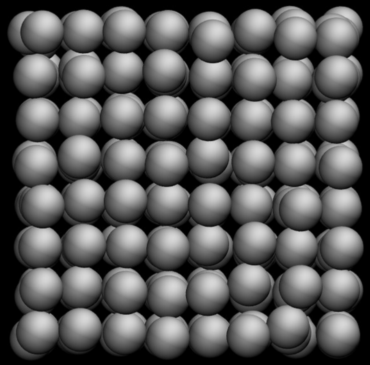


```
units real
atom_style full
bond_style harmonic
angle_style harmonic
pair_style lj/cut/coul/long 10
kspace_style ewald 1e-5
```

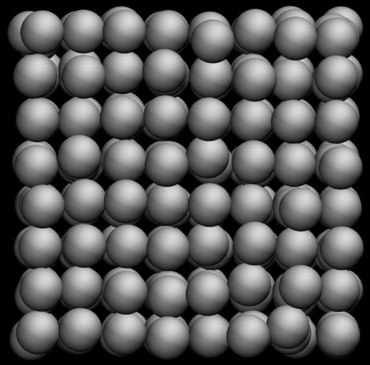
```
region box block -15 15 -15 15 -15 15
  create_box 2 box &
  bond/types 1 &
  angle/types 1 &
  extra/bond/per/atom 2 &
  extra/angle/per/atom 2 &
  extra/special/per/atom 2
```

```
molecule h2omol water.mol
create_atoms 0 random 350 87910 NULL mol h2omol 454756 overlap 1.0 maxtry 50
```

Simulating metal (EAM)

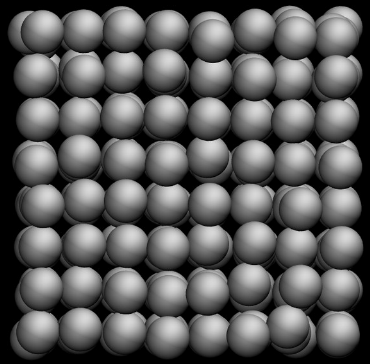


Simulating metal (EAM)



```
# 3) Settings
pair_style eam/alloy
pair_coeff * * Al_zhou.eam.alloy Al
```

Simulating metal (EAM)



```
# 3) Settings
pair_style eam/alloy
pair_coeff * * Al_zhou.eam.alloy Al
```

LAMMPS-GUI - Editor - Al_zhou.eam.alloy

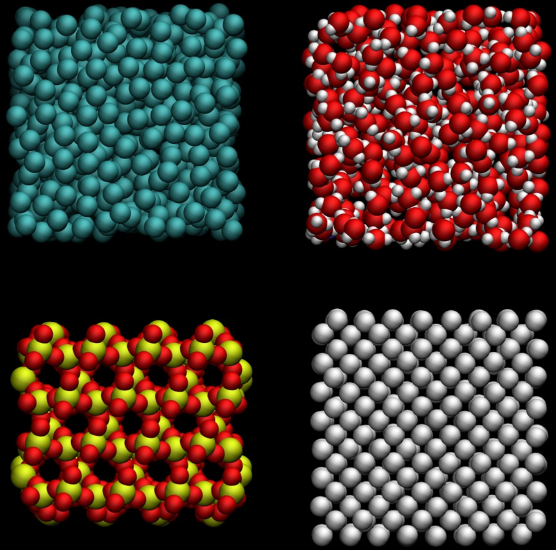
File Edit Run View Tutorials About

```
DATE: 2007-10-12 UNITS: metal CONTRIBUTOR: G. Ziegenhain, gerolf@ziegenhain.com CITATION:
Zhou et al, Acta Mater, 49, 4005 (2001)
#-> LAMMPS Potential File in DYNAMO 86 setfl Format <-#
# Zhou Al Acta mater(2001)49:4005
1 Al
10001 0.00559521603477821424 10001 0.00101014898510148996 10.10250000000000092371
1 26.98199999999999931788 4.04100000000000036948 FCC
0.00000000000000022204
-0.00226922568192955681
-0.00453722203998531502
-0.00680398941276168408
-0.00906952813885751261
```

Explore the repository

https://tinyurl.com/y9w53mrj (zip)

https://tinyurl.com/56mecmb3 (tar.gz)



https://github.com/simongravelle/DIADEM